

Separating phases of allopolyploid evolution with resynthesized and natural *Capsella bursa-pastoris*

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Abstract

Summary

Allopolyploidization is a frequent evolutionary transition in plants that combines whole-genome duplication (WGD) and interspecific hybridization. The genome of an allopolyploid species results from initial interactions between parental genomes and long-term evolution. Telling apart the contributions of these two phases is essential to understanding the evolutionary trajectory of allopolyploid species. Here, we compared phenotypic and transcriptomic changes in natural and resynthesized *Capsella* allotetraploids with their diploid parental species. We focused on phenotypic traits associated with the selfing syndrome and on transcription-level phenomena such as expression level dominance, transgressive expression, and homoeolog expression bias.

We found that selfing syndrome, high pollen and seed quality in natural allotetraploids likely resulted from long-term evolution. Similarly, transgressive expression and most down-regulated expression-level dominance were only found in natural allopolyploids. Natural allotetraploids also had more expression-level dominance toward the self-fertilizing parental species than resynthesized allotetraploids, mirroring the establishment of the selfing syndrome. However, short-term changes mattered, and 40% of the cases of expression-level dominance in natural allotetraploids were already observed in resynthesized allotetraploids. Resynthesized allotetraploids showed striking variation of homoeolog expression bias among chromosomes and individuals. Homoeologous synapsis was its primary source and may still be a source of genetic variation in natural allotetraploids.

In conclusion, both short- and long-term mechanisms contributed to transcriptomic and phenotypic changes in natural allotetraploids. However, the initial gene expression changes were largely reshaped during long-term evolution leading to further morphological changes.

eLife assessment

This **important** study offers new insight into how floral and reproductive phenotypes and gene expression evolve in allopolyploids. The authors marshal **compelling** evidence, using well-constructed genetic lines, RNA sequencing, and phenotypic analyses to distinguish the roles of hybridization, whole genome duplication, and subsequent evolution in phenotypes associated with the selfing syndrome and in gene expression. The work will be of interest to researchers working in plant speciation and genomics, as well as those more broadly interested in the effects of genome copy number on phenotypic and expression evolution.

Introduction

Allopolyploidization is the coupling of whole genome duplication and interspecific hybridization, resulting in organisms possessing two or more diverged genomes. This intriguing evolutionary transition is widespread in nature (Albertin & Marullo, 2012 [↗](#); Barker *et al.*, 2016 [↗](#)), and is of agricultural importance (Behling *et al.*, 2019 [↗](#)). Allopolyploidization is expected to have both short-term and long-term consequences: not only can the merging of divergent genomes itself be seen as a macromutation, but it also triggers subsequent genomic changes over distinct time scales.

Right after allopolyploidization or within a few generations, various genomic and transcriptomic changes can be caused by a series of mechanisms, including DNA methylation repatterning (Edger *et al.*, 2017 [↗](#); Li *et al.*, 2019 [↗](#)), reactivation of transposable elements (TE, reviewed in Vicent & Casacuberta, 2017 [↗](#)), chromosome rearrangements, including homoeologous exchanges (Parisod *et al.*, 2009 [↗](#); Szadkowski *et al.*, 2010 [↗](#); Lashermes *et al.*, 2014 [↗](#); Xiong *et al.*, 2021 [↗](#)), and intergenomic interactions between regulatory elements (Shi *et al.*, 2012 [↗](#); Hu & Wendel, 2019 [↗](#)). These multifaceted effects were initially proposed to be dramatic but are likely smoother and more subtle than initially thought. The fact remains that these short-term mechanisms add further complexity to the genetic variation gathered from parental lineages. Genetic changes can reinforce some initial epigenetic changes, leading to long-term heritable consequences in established allopolyploids. For instance, an epigenetically downregulated/silenced gene copy is more likely to degenerate than the other copy due to weaker purifying selection.

Apart from instant genomic changes, allopolyploidization also alters multiple genetic attributes, impacting the long-term evolution of allopolyploid genomes. First, as a minority cytotype, newly formed allopolyploid populations often experience a bottleneck (Levin, 1975 [↗](#); Novikova *et al.*, 2017 [↗](#); Griffiths *et al.*, 2019 [↗](#)). This bottleneck reduces genetic variation within allopolyploid species and favors the fixation of neutral or slightly deleterious mutations (Novikova *et al.*, 2017 [↗](#)). Second, with an extra genome, allotetraploid species could undergo a period of relaxed purifying selection (Lynch & Conery, 2000 [↗](#); Douglas *et al.*, 2015 [↗](#); Paape *et al.*, 2018 [↗](#)). Relaxed selection also accelerates the accumulation of deleterious mutations on allopolyploid genomes. At the same time, it facilitates neofunctionalization by allowing functional mutations to accumulate in one paralog while maintaining the ancestral function through the second (Ohno, 1970 [↗](#)). Third, allopolyploidization immediately distorts both the relative and absolute dosage of gene product, which further alters physiological balance and efficiency (Anneberg & Segraves, 2020 [↗](#); Yu *et al.*, 2021 [↗](#); Domínguez-Delgado *et al.*, 2021 [↗](#)). In the long term, both relative and absolute dosages of gene expression of allopolyploid genomes are expected to be under selection (Bekaert *et al.*, 2011 [↗](#)), and gradually adapt to a polyploid or hybrid state (Bomblies, 2020 [↗](#)). Under the joint

action of these forces, allopolyploid subgenomes are further co-adapted and degenerated, and subgenomes are often biasedly retained, termed biased fractionation (Schnable *et al.*, 2011; Tang *et al.*, 2012; Renny-Byfield *et al.*, 2015; Wendel *et al.*, 2018).

Both short-term reactions and long-term evolution can generate novel evolutionary opportunities and potentially allow allopolyploid lineages to have advantages in adaptation to novel environments (Baniaga *et al.*, 2020). In established allopolyploids, phenomena caused by long-term evolutionary forces can be confounded by traces of short-term genomic changes. The relative contributions of short- and long-term mechanisms to genomic changes in allopolyploids can be assessed by comparing established natural allopolyploids with resynthesized allopolyploids (e.g., Wang *et al.*, 2006, 2016; Buggs *et al.*, 2011; Yoo *et al.*, 2013; Zhang *et al.*, 2016b).

Variation and novelties in gene expression caused by allopolyploidization are often assessed by homoeolog expression bias (HEB) and non-additive gene expression (Grover *et al.*, 2012; Yoo *et al.*, 2013; Zhang *et al.*, 2016a; Wu *et al.*, 2018; Shan *et al.*, 2020). Homoeolog expression bias measures the separate contributions of gene copies from different parental species (homoeologs) and non-additive gene expression measures the deviation of the total expression of both homoeologs from an intermediate value between parental species. Non-additive patterns of gene expressions are further classified as expression level dominance (ELD) and transgressive expression (TRE). Expression level dominance means that the total expression of both homoeologs is similar to the expression level of only one parental species (Grover *et al.*, 2012), but differs from the expression level of the other. TRE means that gene expression in allopolyploids is higher or lower than in both parental species. Variation of homoeolog expression bias and non-additive gene expression in allopolyploids can be triggered by several mechanisms in the early generations of the new allopolyploid; or alternatively, they may arise during long-term evolution due to either neutral or selective processes.

Capsella bursa-pastoris is a natural allotetraploid plant species which originated about 100,000 years ago (Douglas *et al.*, 2015). Two diploid species, *C. orientalis* and *C. grandiflora*, are extant relatives of the maternal and paternal progenitors (hereinafter referred to as parental species) of *C. bursa-pastoris*, respectively (Hurka *et al.*, 2012; Douglas *et al.*, 2015). *C. grandiflora* is self-incompatible (SI), but *C. orientalis* was already self-compatible (SC) before the formation of *C. bursa-pastoris* (Bachmann *et al.*, 2019). *C. bursa-pastoris* is also an SC species, with typical selfing-syndrome characteristics. In particular, it has smaller petals, fewer pollen grains, and shorter styles than the outcrossing *C. grandiflora* (Neuffer & Paetsch, 2013). Yet, it remains unclear whether the inconspicuous flower phenotypes of *C. bursa-pastoris* only reflect the dominance relationship of the parental alleles or if these traits have also evolved post-allopolyploidization.

Natural *C. bursa-pastoris* exhibits disomic inheritance (Hurka *et al.*, 1989; Roux & Pannell, 2015), with which chromosomes only recombine and segregate with their homologs during meiosis, but not with homoeologs. However, the strictness of disomic inheritance in *C. bursa-pastoris* has not been tested. In general, the two subgenomes of *C. bursa-pastoris* are still well-retained and functional. There is no sign of large-scale gene loss or silencing, although purifying selection has been weaker genome-wide (Douglas *et al.*, 2015), and the *C. orientalis*-derived subgenome (Cbp_co) has accumulated more putatively deleterious mutations than the *C. grandiflora*-derived subgenome (Cbp_cg), both before and after the formation of *C. bursa-pastoris* (Douglas *et al.*, 2015; Kryvokhyzha *et al.*, 2019a). The majority of genes were expressed from both homoeologs, and on average there was only a slight homoeolog expression bias toward Cbp_cg homoeologs (Douglas *et al.*, 2015; Kryvokhyzha *et al.*, 2019a). Most genes are additively expressed in natural *C. bursa-pastoris*, but expression level dominance and transgressive gene expression have also been observed (Kryvokhyzha *et al.*, 2019a). Despite the moderate homoeolog expression bias and non-additive expression, gene expression in *C. bursa-pastoris*

showed some striking tissue-specific features (Kryvokhyzha *et al.*, 2019a [↗](#)). In flowers, gene expression levels in *C. bursa-pastoris* resembled those in *C. orientalis*, while in leaves and roots, gene expression levels were more similar to those in *C. grandiflora*.

In contrast to the drastic genomic or transcriptomic changes observed in allopolyploid wheat (Zhang *et al.*, 2016a [↗](#)), *Brassica* (Szadkowski *et al.*, 2010 [↗](#); Lloyd *et al.*, 2018 [↗](#)), and *Tragopogon* (Chester *et al.*, 2012 [↗](#)), natural *C. bursa-pastoris* represents another paradigm where established allopolyploid species show only mild genomic changes and expression bias. This contrast raises questions. Was the genome of natural *C. bursa-pastoris* less affected by putative short-term mechanisms, or was it the result of 100,000 years' evolution, which filtered out or compensated for the initial drastic changes? What are the relative strengths of short-term mechanisms and long-term evolution in shaping genomic and phenotypic variation in allopolyploids?

Resynthesized allopolyploids are the closest approximation to the early stage of natural allopolyploids. They provide a reference point for separating the short-term effects of allopolyploidization from long-term evolutionary changes. The present study builds upon Duan *et al.* (2023) [↗](#), which showed that hybridization played a much larger role than whole genome doubling during the creation of resynthesized polyploids in the *Capsella* genus. Here, we compared transcriptomes and phenotypes of resynthesized *C. bursa-pastoris*-like allotetraploids with natural *C. bursa-pastoris* and its two diploid progenitors. We focused on teasing apart the contributions from short- and long-term processes to 1) phenotypes, 2) non-additive gene expression, and 3) homoeolog expression bias in *Capsella* allotetraploids.

Results

The selfing syndrome was observed in natural *C. bursa-pastoris* but not in resynthesized allotetraploids

The breakdown of self-incompatibility in allotetraploid *Capsella* can directly result from hybridizing with the self-fertilizing species (Bachmann *et al.*, 2021 [↗](#); Duan *et al.*, 2023 [↗](#)). We explored to what extent the development of a selfing syndrome was instantly achieved after allopolyploidization or, instead, developed later on by comparing phenotypes of resynthesized allotetraploids (groups Sd and Sh), natural *C. bursa-pastoris* (Cbp) and the diploid parental species, *C. grandiflora* (Cg2) and *C. orientalis* (Co2). The resynthesized allotetraploids were generated with individuals from one population of each diploid parental species (Figure 1 [↗](#)), and the Sd (“WGD-first”) and Sh (“hybridization-first”) groups only differed in the order of WGD and hybridization (Duan *et al.*, 2023 [↗](#)). There were six “lines” in each of the five plant groups. For the Sd and Sh groups, each line represented an independent allopolyploidization event, while the six lines of natural *C. bursa-pastoris* were from six different populations (Figure 1c [↗](#) and Figure 1 [↗](#)–Source Data 1), representing three major genetic clusters of the wild *C. bursa-pastoris* (Kryvokhyzha *et al.*, 2016 [↗](#), 2019b [↗](#)). For diploid parental groups, a line referred to a full-sibling family resulting from self-fertilization (Co2) or one controlled cross (Cg2). Phenotypes of the five groups were measured in a growth chamber on about 36 individuals per plant group (6 individuals×6 lines).

Resynthesized and natural allotetraploids had distinct floral morphologies (Figure 2a–g [↗](#)). Indeed, natural allotetraploids had significantly shorter and narrower petals, sepals and pistils and shorter stamina than resynthesized allotetraploids (one-way ANOVA, $F_{4,160} > 78$ and $p < 0.001$ in all seven tests; Tukey's HSD test, $\alpha = 0.01$). Pollen and seed production was also affected. Natural allotetraploids had fewer pollen grains per flower (Figure 2i [↗](#)). While the number of pollen grains in resynthesized allotetraploids was intermediate between the two parental species, the number of pollen grains of the natural allotetraploid group was now similar to that of the Co2 group (one-way ANOVA, $F_{4,137} = 164.6$, $p < 0.001$; Tukey's HSD test, $\alpha = 0.01$). Moreover, the number of seeds per fruit in natural allotetraploids was much larger than in resynthesized allotetraploids.

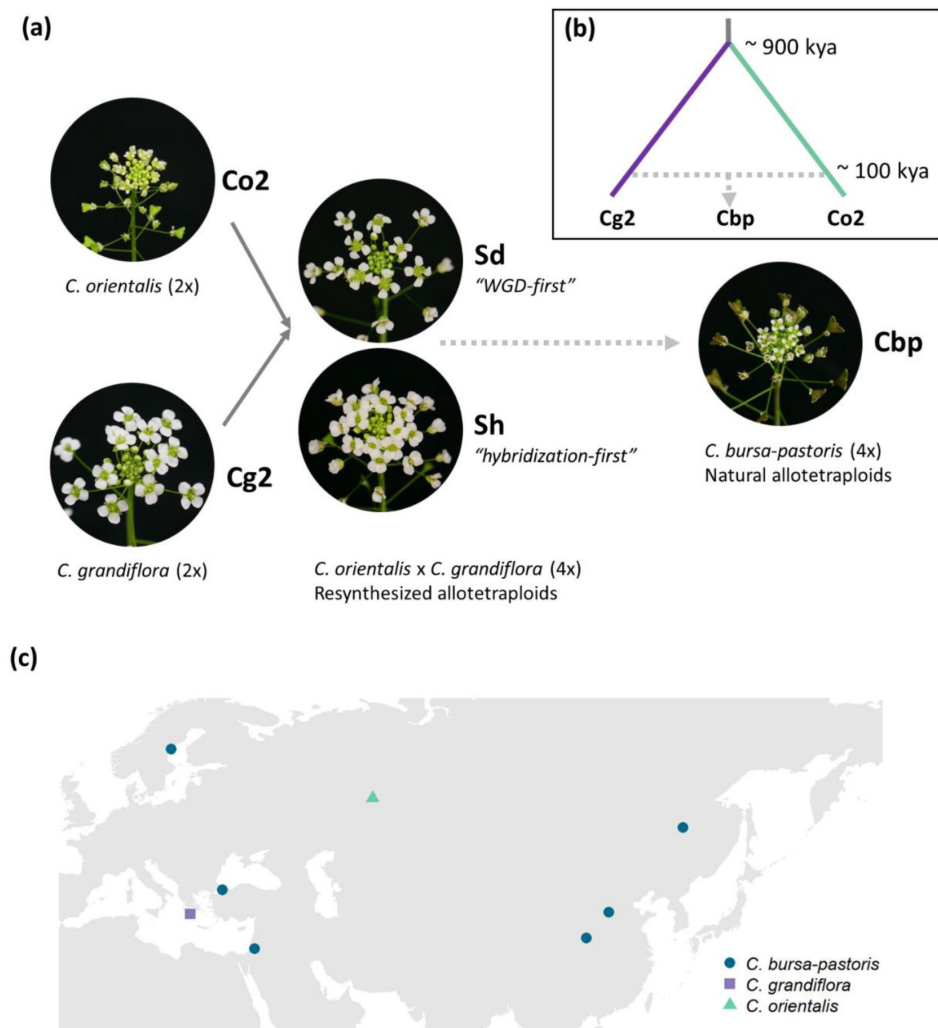


Figure 1

Plant material used in the present study. (a) Five groups of *Capsella* plants. Diploid species (Co2 and Cg2 groups) and the second generation of resynthesized allotetraploids (Sd and Sh groups) were from Duan *et al.* (2023) [DOI:10.1101/2023.03.01.530000](#). Samples of natural allotetraploids, *C. bursa-pastoris*, were added to the present study. (b) Phylogenetic relationship of the three natural species used in the present study, modified from Douglas *et al.*, (2015) [DOI:10.1111/1365-3113.12500](#); *C. bursa-pastoris* originated from the hybridization between the ancestral population of *C. orientalis* and the (*C. grandiflora* + *C. rubella*) lineage, and *C. rubella* were omitted from the figure; kya: thousand years ago. (c) Geographic origin of the *Capsella* samples.

The resynthesized allotetraploid groups had a similar number of seeds in 10 fruits to that of the Cg2 group, whereas the number of seeds in 10 fruits in the natural allotetraploid group was even higher than that of the Co2 group (**Figure 2j** [↗](#); one-way ANOVA, $F_{4,146} = 152.5$, $p < 0.001$; Tukey's HSD test, $\alpha = 0.01$).

The architecture and phenology of the whole plant were affected too. The stem length of natural Cbp was shorter than in resynthesized allotetraploids but was similar to the stem length of the Co2 group (**Figure 2h** [↗](#); one-way ANOVA, $F_{4,166} = 84.5$, $p < 0.001$; Tukey's HSD test, $\alpha = 0.01$). Finally, plants of the Cbp group flowered earlier than those of the Sd group, but at a similar time as those of the Sh group (**Figure 2k** [↗](#); one-way ANOVA with hc3 White's correction, $F_{4,165} = 49.2$, $p < 0.001$; Tukey's HSD test, $\alpha = 0.01$).

Pollen viability and seed quality improved in natural *Capsella* allotetraploids

Pollen viability and the proportion of normal seeds were compared between resynthesized and natural allotetraploids. For both traits, we observed a decrease in pollen viability in resynthesized allotetraploids followed by recovery in natural Cbp. Both Sd and Sh groups had lower proportions of viable pollen than the diploid parental species, but the proportion of viable pollen in natural Cbp was similar to that of the diploid parental species (**Figure 2l** [↗](#); GLM, quasi binomial, $F_{4,137} = 24.4$, $p < 0.001$; Tukey's HSD test, $\alpha = 0.01$). The resynthesized allotetraploids generated a higher proportion of abnormal seeds than the three natural species (**Figure 2m** [↗](#); GLM, quasi binomial, $F_{4,146} = 59.2$, $p < 0.001$; Tukey's HSD test, $\alpha = 0.01$). The average percentage of normal seeds was $69.6 \pm 4.3\%$ in the Sd group and $77.5 \pm 2.2\%$ in the Sh group. In contrast, the natural Cbp had almost no abnormal seeds, with a percentage of normal seeds of $99.6 \pm 0.8\%$.

A two-step evolution of the global expression pattern of natural allopolyploid Cbp

To compare the gene expression pattern of the five plant groups, RNA-sequencing was conducted for one individual per line and six lines per group, using young inflorescences (flowers) and leaves, respectively. Expression levels were determined for 21,937 genes in flower samples and 18,999 genes in leaf samples after excluding genes with CPM > 1 in less than two samples. The overall gene expression pattern was visualized with multi-dimensional scaling (MDS) analysis (**Figure 3a,b** [↗](#)). For unphased gene expression, the resynthesized allotetraploids lay between the diploid parental species in both flowers and leaves. Natural Cbp samples were also intermediate between parental species in the first dimension but were far from the resynthesized allotetraploids in the other dimension, showing the effect of long-term evolution in Cbp and possibly also the divergence between extant diploid species and the real progenitors.

The expression levels of separate homoeologs in allotetraploids were determined with the diagnostic SNPs between the two diploid species. For the Sd, Sh, and Cbp groups, 52.8%, 53.7%, and 44.7% of the mapped reads could be assigned to one of the homoeologs, respectively. The expression pattern of each allotetraploid subgenome was more similar to the corresponding diploid progenitor (**Figure 3c,d** [↗](#)). The pattern of expression of resynthesized allotetraploids was intermediate between those of diploid progenitors and natural Cbp.

Differential expression analysis was performed among the five plant groups, using the down-sampled unphased gene expression data. With a threshold of $FC > 2$ and $FDR < 0.05$, no significant differentially expressed genes (DEGs) were found between the two resynthesized allotetraploid groups, while 311 to 2888 DEGs were revealed in other group contrasts (Supplementary files 3,4). There are two salient features. First, compared to either diploid progenitor, most DEGs in both resynthesized and natural allotetraploid groups were up-regulated (Supplementary file 4). The

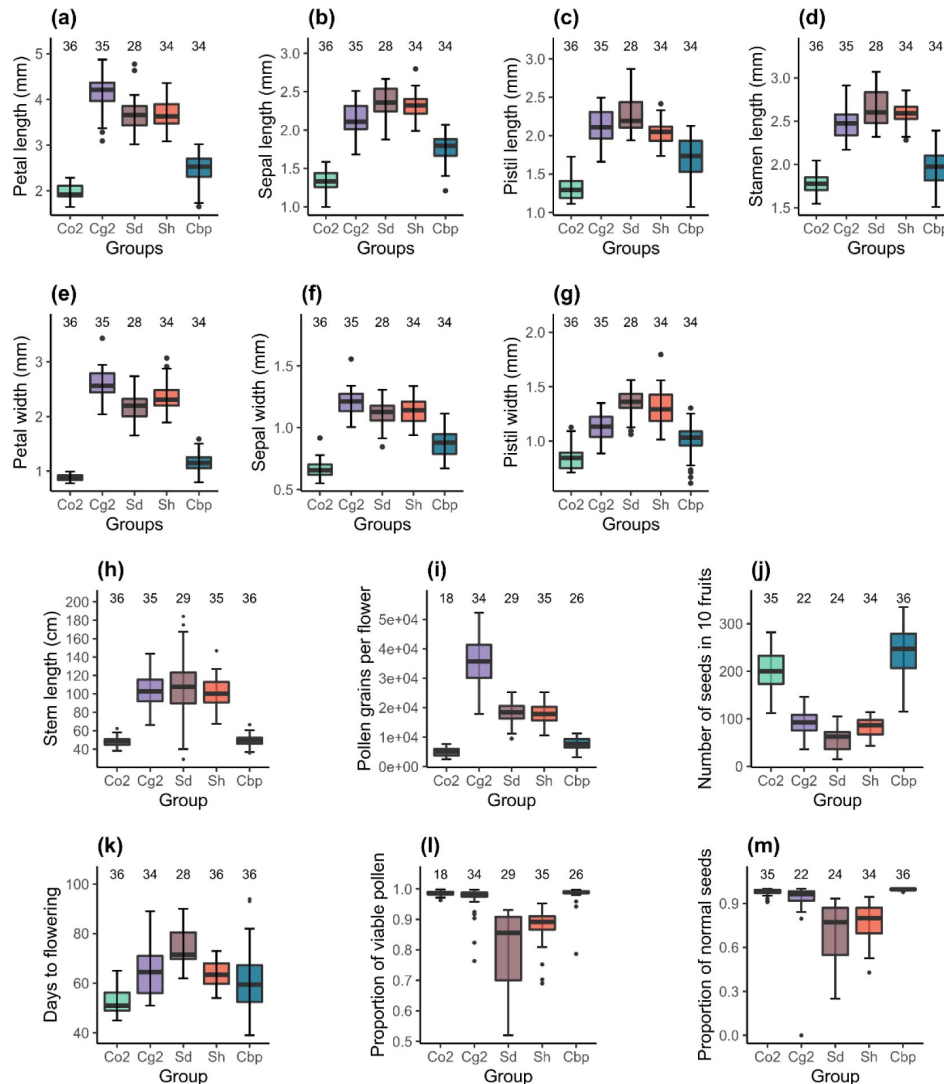


Figure 2

Phenotypic traits of the five *Capsella* groups. Co2: diploid *C. orientalis*; Cg2: diploid *C. grandiflora*; Sd: whole-genome-duplication-first resynthesized allotetraploids; Sh: hybridization-first resynthesized allotetraploids; Cbp: natural allotetraploid *C. bursa-pastoris*. The measured traits were (a) petal length, (b) sepal length, (c) pistil length, (d) stamen length, (e) petal width, (f) sepal width, (g) pistil width, (h) length of the longest stem, (i) number of pollen grains per flower, (j) number of seeds in ten fruits, (k) number of days from germination to the opening of the first flower, (l) proportion of viable pollen grains, and (m) proportion of normal seeds in ten fruits.

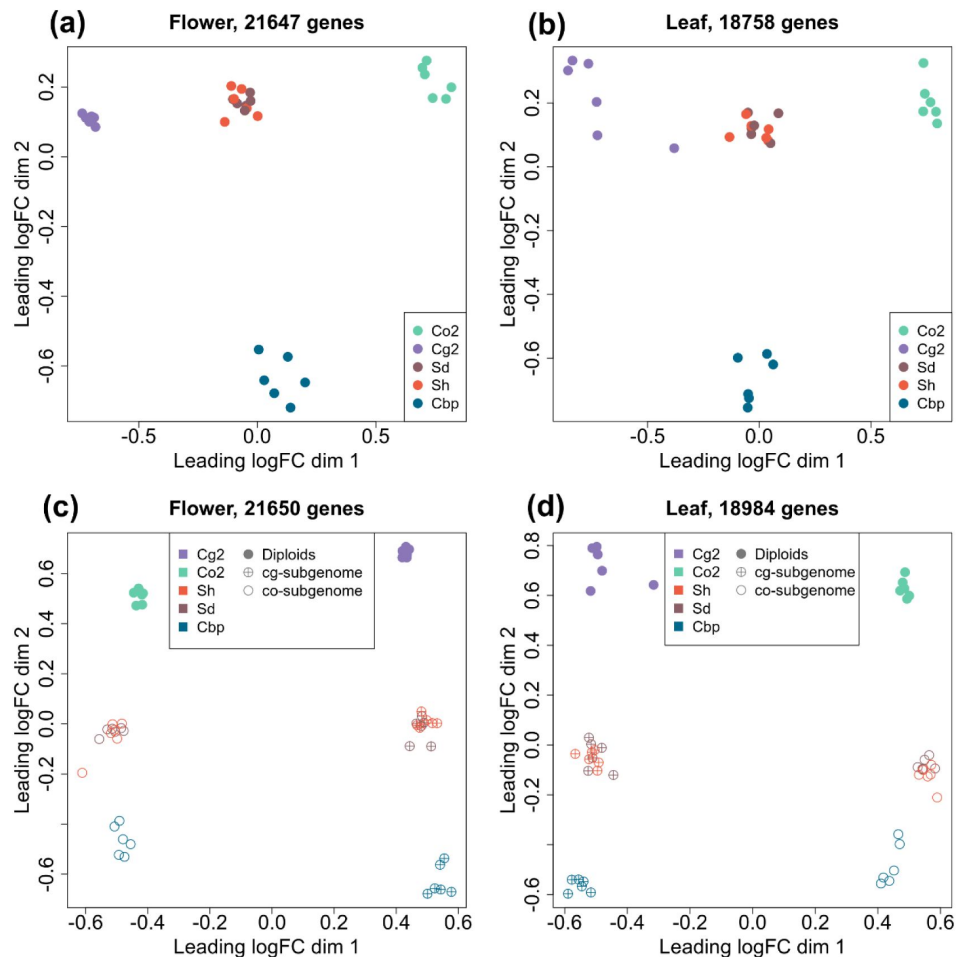


Figure 3

Multi-dimensional scaling (MDS) analyses of gene or homoeolog expression in two tissues. Down-sampled gene expressions were used to compare gene expression patterns of the five plant groups in either flowers (a) or leaves (b). The five groups of *Capsella* plants are: diploid parental species *C. grandiflora* (Cg2) and *C. orientalis* (Co2), hybridization-first (Sh) and whole-genome-duplication-first (Sd) resynthesized allotetraploids, and natural allotetraploid *C. bursa-pastoris* (Cbp). Separated homoeolog expressions in allotetraploids were then compared with rescaled gene expression of diploid groups in both flowers (c) and leaves (d). All the MDS analyses used genes with count-per-million (CPM) > 1 in at least two samples, and expression levels were normalized with the trimmed mean of M-values (TMM) method.

proportion of down-regulated DEGs increased, nevertheless, in natural allotetraploids. Second, both resynthesized allotetraploids, Sd and Sh, have much more DEG with Co2 than with Cg2. However, this is no longer the case in Cbp where the two comparisons yielded similar results.

Although we could not make a clear expectation for gene ontology (GO) terms that would be overrepresented in DEGs between resynthesized and natural allopolyploids, we are not the only study that compared newly resynthesized and established allopolyploids, and GO terms that were repeatedly revealed by exploratory analysis may give a hint for future studies. For this reason, we further performed a GO enrichment analysis, using genes that were differentially expressed in both Cbp-Sd and Cbp-Sh contrasts as the test set, and all expressed genes with GO annotations as the background set. In flowers or leaves, the top ten most-enriched GO terms for biological processes were related to proteolysis, xenobiotic transport, regulation of translational fidelity, telomere maintenance and organization, DNA geometric change and duplex unwinding, cellular response to toxic substance, aminoacylation, peptidyl-lysine methylation, and heterochromatin formation and organization (Supplementary file 5). But after adjusting p-values with the Benjamini-Hochberg procedure, only GO term proteolysis (GO:0006508) was significantly overrepresented in DEGs between resynthesized and natural allotetraploids in leaves (Fisher's exact test, adjusted p-value=0.0175).

Both short- and long-term mechanisms contributed to expression level dominance in natural Cbp, but transgressive expressions were mainly from long-term evolution

Non-additive gene expression shared by natural allotetraploids may be triggered right after allopolyploidization by short-term deterministic mechanisms, such as intergenomic interactions of regulatory elements. Alternatively, non-additive expression may have been caused by mechanisms with stochastic effects or arose later during long-term evolution. We explored to what extent the non-additive expression shared by natural allotetraploids could reflect short-term deterministic mechanisms. By comparing gene expression levels in allotetraploids and diploid species, 21,647 genes in flowers and 18,758 genes in leaves were classified into one of the ten expression categories, using the results of DE analysis on unphased gene expression ($FC > 2$ and $FDR < 0.05$). We focused on complete expression level dominance and TREs: complete expression level dominance is obtained when the gene expression level in an allopolyploid group is similar to that in one diploid group but not to the expression in the other diploid group, and TRE is detected when the gene expression level in an allopolyploid group is either higher or lower than in both diploid groups.

The percentage of genes showing complete expression level dominance was altogether limited but doubled in natural allotetraploids relative to resynthesized allotetraploids (5.5% of genes in resynthesized allotetraploids and 10.2% in natural allotetraploids. **Figure 4a,b** and **Figure 4c**—Source Data 1). Genes with expression level dominance and the directions of expression level dominance were highly shared between the two resynthesized allotetraploid groups (**Figure 4c**, **Figure 4**—figure supplement 1). The majority of these shared expression level dominance were retained in natural allotetraploids (63.3% in flowers and 72.2% in leaves), suggesting that short-term deterministic mechanism contributed to expression level dominance in natural allotetraploids. However, Cbp-specific expression level dominance was also abundant, comprising more than half of the cases found in natural allotetraploids (56.6% in flowers and 60.8% in leaves), thereby showing the effects of long-term evolution.

The direction of expression level dominance shifted between the resynthesized and natural allotetraploids (**Figure 4a,b** and **Figure 4c**—Source Data 1). In resynthesized allotetraploids, most cases of expression level dominance were up-regulated, and the number of genes with expression level dominance toward *C. grandiflora* (Cg-ELD) was about twice of that toward *C. orientalis* (Co-ELD). Natural allotetraploids still had more up-regulated expression level dominance

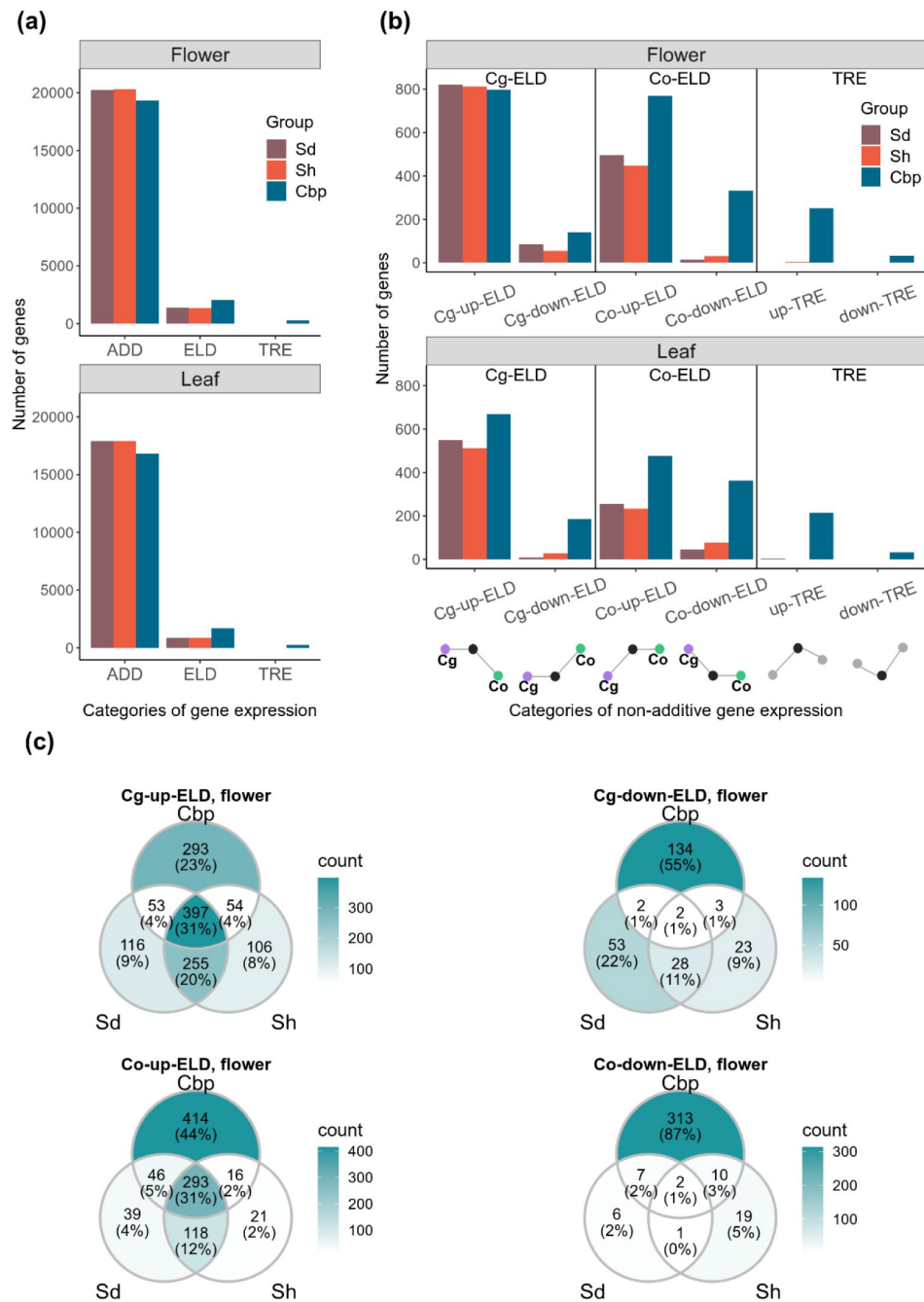


Figure 4

Additive and non-additive expression in allotetraploid groups. Sd: whole-genome-duplication-first resynthesized allotetraploids; Sh: hybridization-first resynthesized allotetraploids; Cbp: natural allotetraploid *C. bursa-pastoris*. a) Number of genes that showed additive expression (ADD, including partial expression level dominance), complete expression level dominance (ELD), and transgressive expression (TRE) in each allotetraploid group. b) Genes with complete expression level dominance or transgressive expression were further classified by whether they were up- or down-regulated in allotetraploids, and whether the expression level was similar to *C. grandiflora* (Cg-ELD) or *C. orientalis* (Co-ELD). c) Venn diagram of genes with complete expression level dominance of the three allotetraploid groups in flowers, separated by directions of expression level dominance.

than down-regulated ones, but the proportion of down-regulated cases increased. The proportion of Co-ELD had also increased in natural allotetraploids. In flowers, natural allotetraploids had more Co-ELD (1101) than Cg-ELD (938), and the number of Cg- and Co-ELDs were similar in leaves (Cg-ELD: 854, Co-ELD: 839).

Almost no TRE was found in resynthesized allotetraploids (less than five genes in either Sd or Sh group and in either tissue, **Figure 4a,b**, and **Figure 4**—Source Data 1). In contrast, about 1.3% of genes in Cbp showed TRE in both flowers and leaves.

Segregation and recombination of homoeologous chromosomes were a major source of homoeolog expression bias variation in resynthesized *Capsella* allotetraploids

Homoeolog expression bias of genes in an allotetraploid individual was measured by the ratio of expression of the *C. grandiflora*-origin homoeolog (cg) to the total expression of both homoeologs (cg/(cg+co)). To obtain a reliable gene expression ratio, lowly expressed genes (CPM(cg+co) < 1 in any allotetraploid individual) were excluded from this analysis. Eventually, the homoeolog expression ratio was calculated for 18,255 genes in flowers, and 15,581 genes in leaves.

Overall, none of the three allotetraploid groups showed a strong average homoeolog expression bias. In flowers, the average homoeolog expression bias was 0.499 ± 0.001 , 0.531 ± 0.001 , and 0.475 ± 0.001 for the Sd, Sh, and Cbp groups, respectively. In leaves, the average homoeolog expression bias was 0.504 ± 0.001 , 0.532 ± 0.001 , and 0.477 for the Sd, Sh, and Cbp groups. When averaged among individuals, the homoeolog expression bias of resynthesized allotetraploids had smaller gene-wise variation than that of the Cbp group (Levene's test, p-value < 0.001).

Among resynthesized allotetraploids, although the average homoeolog expression ratio was not systematically biased toward Cg or Co, homoeolog expression bias showed great variation among chromosomes and individuals (**Figure 5a,b**, **Figure 5**—figure supplements 1 and 2). The distribution of homoeolog expression bias in some chromosomes had peaks around 0, 0.25, 0.75, or 1, but the shape of the distribution was almost identical between flower and leaf samples. When homoeolog expression bias of genes was plotted along chromosomal positions, we found that the extra peaks in the distribution of homoeolog expression bias can be further explained by large genomic segments separated by a sudden change of average homoeolog expression bias (**Figure 5c**, **Figure 5**—figure supplements 3-6). Altogether, the pattern suggested that some chromosomes or chromosomal regions in resynthesized allotetraploids had an unbalanced number of cg- and co-homoeologs (not 2:2), which were likely caused by the segregation and recombination of homoeologous chromosomes. Both the segregation and recombination of homoeologous chromosomes are outcomes of homoeologous synapsis (synapsis between homoeologous chromosomes during meiosis), which reflects polysomic or mixed inheritance in resynthesized allopolyploids. For short, we refer to both segregation and recombination of homoeologous chromosomes as homoeologous synapsis.

The effect of possessing an unbalanced number of homoeologs largely increased the variation of homoeolog expression bias in resynthesized allotetraploids. The breakpoint between segments with distinct average homoeolog expression bias and the copy number of cg-homoeolog on each segment were estimated with a five-state Hidden Markov Model (HMM), using homoeolog expression bias along chromosomes (**Figure 5c**, **Figure 5**—figure supplement 7). Among the 96 chromosome quartets (two pairs of homologous chromosomes) from the 12 resynthesized allotetraploid individuals, only 39 chromosome quartets showed no sign of homoeologous synapsis, i.e., no breakpoint was identified and the estimated number of cg-homoeolog across the chromosome was two. On average 0.833 ± 0.097 (mean $\pm$ se) breakpoint was identified for each chromosome quartet, and 31.0% of genes were estimated to have different numbers of cg- and co-

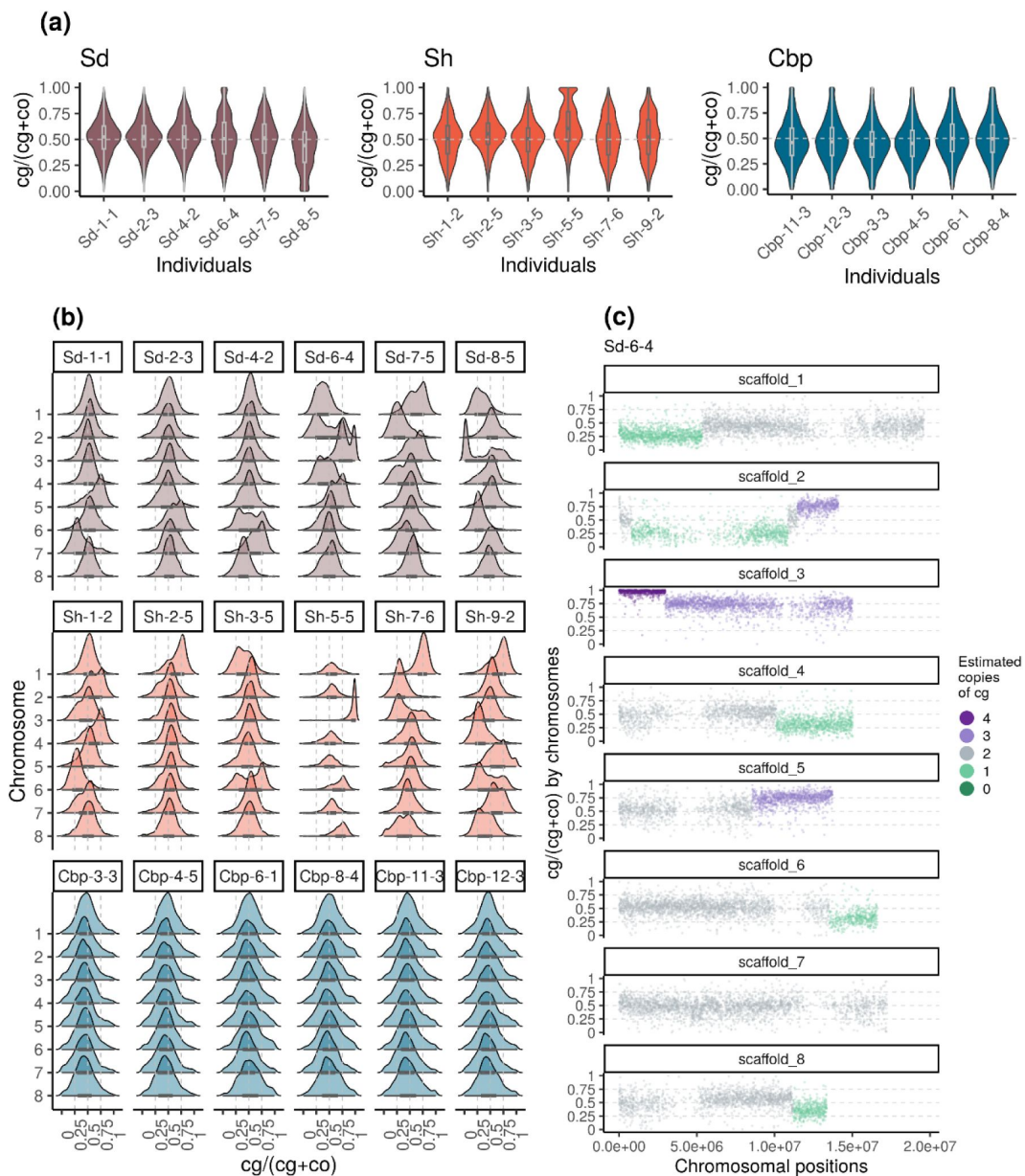


Figure 5

Variation of homoeolog expression bias of the three allotetraploid groups in flowers. Sd: whole-genome-duplication-first resynthesized allotetraploids; Sh: hybridization-first resynthesized allotetraploids; Cbp: natural allotetraploid *C. bursa-pastoris*. Gene-wise homoeolog expression bias was calculated as the expression level of cg-homoeolog divided by the total expression level of both cg- and co-homoeologs ($cg/(cg+co)$). For each individual, homoeolog expression bias was calculated for 18,255 genes, which had count-per-million > 1 in all flower samples. The distribution of homoeolog expression bias was shown by (a) individuals and (b) chromosomes. (c) Homoeolog expression bias was also plotted along chromosome positions to show the sudden change of average homoeolog expression bias between genomic blocks, taking individual Sd-6-4 as an example. The number of cg-homoeologs at each gene estimated by the five-state Hidden Markov Model (HMM) was indicated by five colors. Dark green, light green, grey, light purple and dark purple represent (0, 1, 2, 3, 4) cg-homoeologs and (4, 3, 2, 1, 0) co-homoeologs, respectively.

homoeologs. Finally, for resynthesized allotetraploids, the estimated copy number of homoeologs was able to explain 48.4% and 46.8% of the variance of homoeolog expression bias in flowers and leaves, respectively (GLM with quasi-binomial error distribution, $p < 0.001$ in both tissues).

In contrast to resynthesized allotetraploids, the distribution of homoeolog expression bias of natural Cbp was similar among individuals and chromosomes (**Figure 5a,b**, **Figure 5**—figure supplement 2), although the distribution of homoeolog expression bias of some chromosomes of Cbp also showed weak bumps around 0, 0.25, 0.75 or 1 (**Figure 5b**, **Figure 5**—figure supplement 2). We could not confidently estimate the number of homoeologs or the breakpoint of segments for Cbp with only RNA-sequencing data, as segments resulting from homoeologous exchanges could be shorter in natural Cbp, and the signals of copy number could be blurred by the variance of regulatory divergence. Nevertheless, the HMM segmentation algorithm also identified some candidate segments of which the average homoeolog expression bias strongly deviated from 0.5. Some candidate segments were only shared by individuals from the same population (**Figure 6**, **Figure 6**—figure supplement 1 and 2).

Most resynthesized allotetraploids had less homoeolog expression loss than natural Cbp, but with extreme outliers

Loss of homoeolog expression is a common phenomenon in allopolyploid species, which can be caused by homoeolog loss or silencing (Buggs *et al.*, 2009; Cox *et al.*, 2014; Lashermes *et al.*, 2016). Loss of homoeolog expression may quickly arise after allopolyploidy, or alternatively, reflect a gradual biased gene fractionation during diploidization. We compared the extent of loss of homoeolog expression between resynthesized and natural *Capsella* allotetraploids, and among allotetraploid individuals.

The loss of homoeolog expression was identified from genes with medium to high expression levels in all individuals of the corresponding diploid species (**Figure 7**) to reduce noise from RNA-sequencing and phasing. On average, only 1.0 % of these genes showed homoeolog-specific expression loss in natural Cbp. Most resynthesized allotetraploids have a lower level of homoeolog-specific expression loss than natural Cbp, but three individuals (Sd-6-4, Sd-8-5, and Sh-5-5) showed an extremely high level of homoeolog expression loss. The striking homoeolog expression loss in these three resynthesized allotetraploids was most likely caused by the segregation and recombination of homoeologous chromosomes, as the extreme homoeolog expression bias in the three outliers was restricted to chromosome 3, where the entire chromosome or a large chunk of the chromosome has only expression from one homoeolog (**Figure 5b**, **Figure 5**—figure supplements 3-6).

Expression level dominance is caused by different mechanisms in resynthesized and natural allotetraploids

As homoeologous synapsis seemed to be a major cause of homoeolog expression bias and homoeolog-specific expression loss in resynthesized allotetraploids, we assessed whether it could have also played a role in the evolution of expression level dominance. To do so, we explored the mechanism of expression level dominance in resynthesized allotetraploids by comparing the gene expression change of separate homoeologs relative to the corresponding gene in diploid groups ($\log_2\text{FC}(\text{cg}/\text{Cg}2)$ and $\log_2\text{FC}(\text{co}/\text{Co}2)$) among non-additive gene expression categories (**Figure 8**).

For expression level dominance in resynthesized allotetraploids, different non-exclusive short-term mechanisms would produce different patterns of average expression change of EL-dominant (homoeolog derived from the diploid progenitor to which the total expression of both homoeologs was similar) and EL-recessive homoeologs (the opposite homoeolog), among genes with significant expression level dominance: i) If expression level dominance was mainly caused by possessing

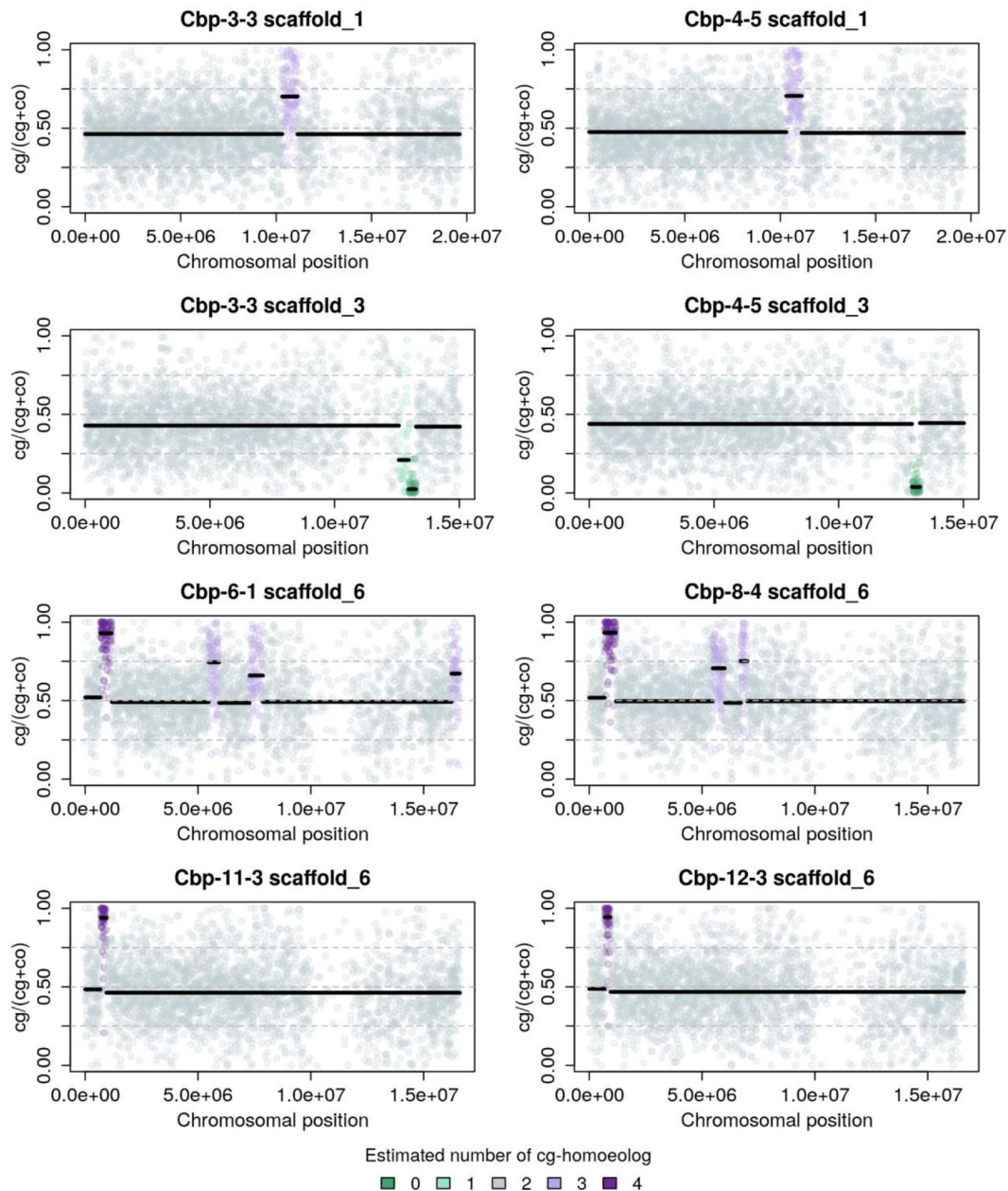


Figure 6

Homoeolog expression bias ($cg/(cg+co)$) along chromosomes of natural allotetraploid *C. bursa-pastoris* in flowers, taking four pairs of chromosome quartets with typical patterns as an example. The number of cg-homoeologs estimated by the five-state Hidden Markov Model was indicated by five colors. The two chromosome quartets in the same row are from the two individuals of the same major genetic cluster of natural *C. bursa-pastoris* (Kryvokhyzha *et al.*, 2016 [DOI](#), 2019b [DOI](#)), showing that some estimated segments with an unbalanced number of cg- and co-homoeologs were shared between the individuals from the same genetic cluster.

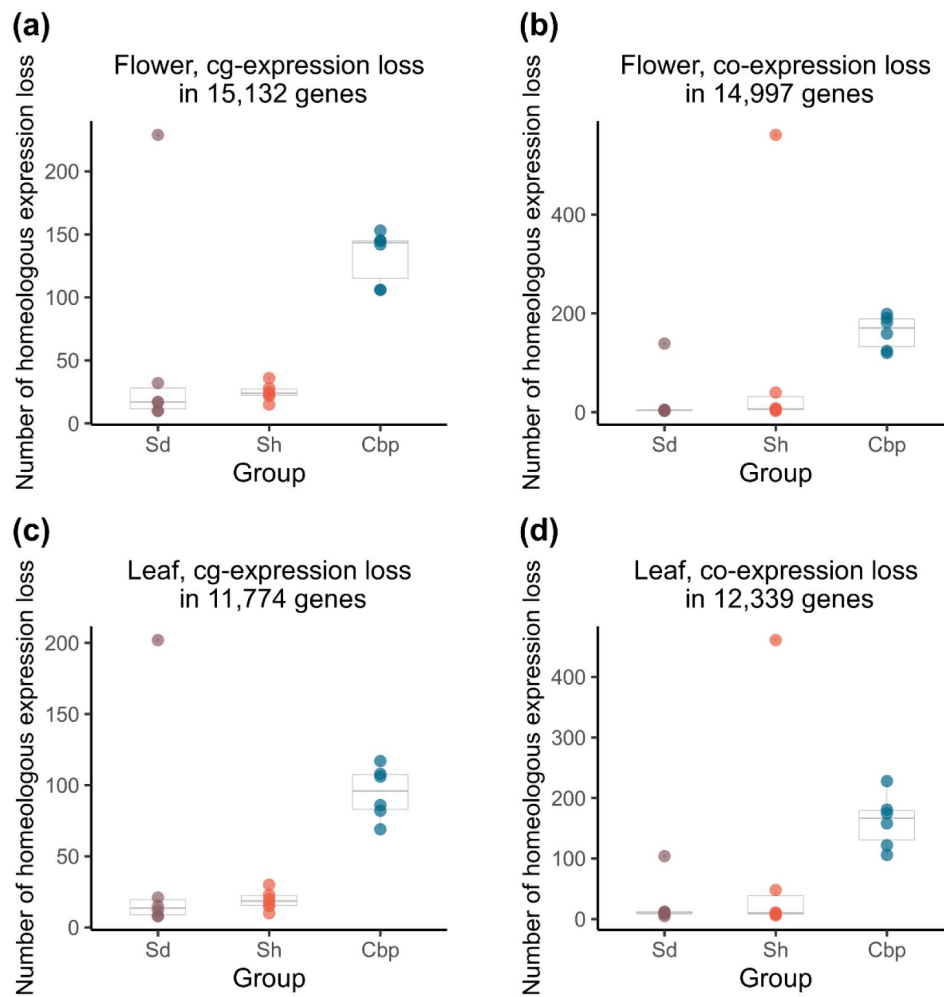


Figure 7

Loss of homeolog expression in resynthesized (Sd and Sh) and natural allotetraploids (Cbp). The number of genes with the expression loss of *C. orientalis*-homeolog (a,c) or *C. grandiflora*-homeolog (b,d) per individual was compared among the three groups of allotetraploids in flowers (a,b) or leaves (c,d). Homeologous genes that had obvious expression (count per million > 5) in all individuals of the corresponding diploid species but almost no expression (count per million < 0.5) in one allotetraploid individual were considered cases of homeolog expression loss.

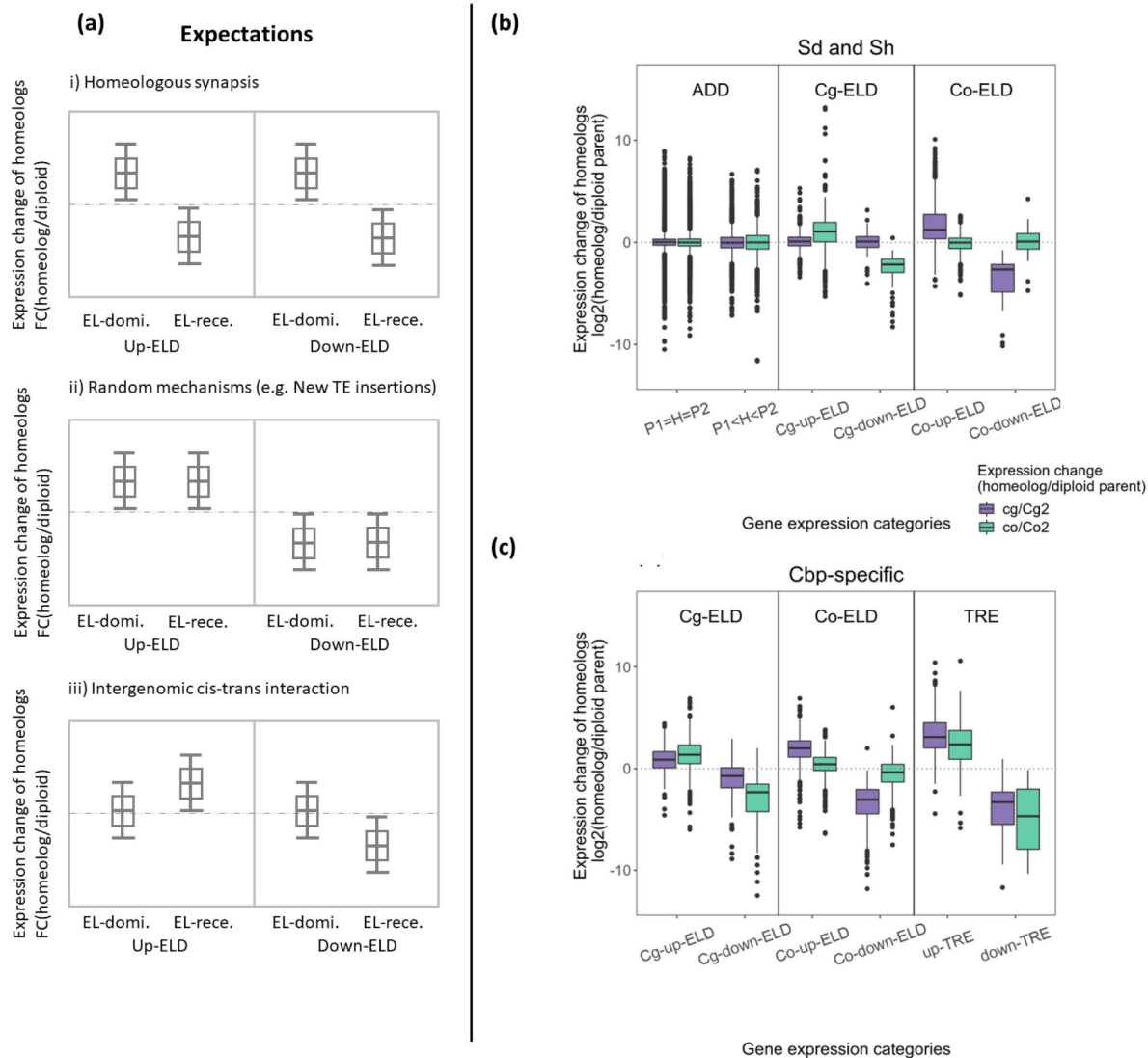


Figure 8

Relationships between homeolog expression change and non-additive gene expression in flowers. (a) Expected patterns of homeolog expression change in each scenario that may explain the cause of expression level dominance (ELD) in resynthesized allotetraploids. (b) Observed homeolog expression change among genes with expression level dominance in resynthesized allotetraploids (Sd and Sh). (c) Observed homeolog expression change among genes with Cbp-specific non-additive expression.

more than two copies of the EL-dominant homoeolog (due to the segregation or recombination of homoeologous chromosomes), we would expect on average the expression of EL-dominant homoeolog to increase, and EL-recessive homoeolog to decrease, in both up-down-regulated cases of expression level dominance. ii) If expression level dominance was mainly caused by mechanisms with random effects, such as TE transposition, on average, there should be no large difference between the expression changes of EL-dominant and EL-recessive homoeologs. Because the occurrence and regulatory effects of new TE transpositions do not depend on the original relative expression level of the two homoeologs, i.e., the highly and lowly expressed homoeologs are equally likely to be up- or down-regulated by new TE transpositions. iii) Predictions for new intergenomic interactions of regulatory elements can be complex, but under a simple scenario (Hu & Wendel, 2019 [\[1\]](#)), expression level dominance may be caused by divergent trans-acting factors. In allopolyploids, stronger trans-acting factors act on the cis-regulatory elements of the opposite homoeolog, causing a similar regulatory effect if transcription rate were not limited by the concentration of these trans-regulators. If this mechanism were the main cause of expression level dominance, on average the EL-recessive homoeolog should have the larger expression change in all categories of expression level dominance, while the expression of EL-dominant homoeolog is not expected to change. In all other cases, we would not have direct inference on the exact mechanism of expression level dominance, but at least the three mechanisms listed above could not be the predominant cause of expression level dominance.

Concerning the expression level dominance found in resynthesized allotetraploids, the change of homoeolog expression fits the third scenario. In all four categories of expression level dominance, the EL-recessive homoeolog had a larger average expression change in the same direction as expression level dominance, while the average expression change of EL-dominant homoeolog was closer to zero (**Figure 8** [\[1\]](#), **Figure 8** [\[2\]](#)—figure supplement 1, and **Figure 8** [\[3\]](#)—Source Data 1).

The Cbp-specific expression level dominance showed a different pattern. Although the EL-recessive homoeolog still had a larger expression change, the EL-dominant also showed non-zero average expression change toward the direction of expression level dominance, especially in Cg-ELD genes. For genes with Cbp-specific TRE, both homoeologs had expression change in the same direction of TRE (**Figure 8** [\[1\]](#), **Figure 8** [\[2\]](#)—figure supplement 1, and **Figure 8** [\[3\]](#)—Source Data 1).

Discussion

Distinguishing parental legacy from the effects of evolutionary forces is essential for interpreting the outcome of allopolyploidization. The short-term genomic interactions in allopolyploids reflect the divergence of parental genomes (Johnson, 2010 [\[1\]](#)). In this sense, short-term transcriptomic changes in new allopolyploids are also part of parental legacy but are not observable with only information from diploid parental species. In this study, we used resynthesized *Capsella* allotetraploids as an approximation of the early stage of the natural allotetraploid species to separate and compare the short- and long-term transcriptomic and phenotypic changes. The timing and pattern of the variation also provided hints for locating the exact mechanism.

The extant diploid species were not the exact parental populations of natural *C. bursa-pastoris*, and the sampled diploid individuals were genetically closer to the resynthesized allotetraploids than to natural *C. bursa-pastoris*, potentially leading to an overestimation of the contribution of long-term mechanisms. However, the divergence between *C. grandiflora* and *C. orientalis* was much more ancient than the formation of *C. bursa-pastoris* (Douglas *et al.*, 2015 [\[1\]](#)). Therefore, the molecular divergence between *C. grandiflora* and *C. orientalis* after the formation of *C. bursa-pastoris* is only a small fraction of the total divergence.

Besides, the mating system of the real parental populations of *C. bursa-pastoris* was likely the same as today: A nonfunctional self-incompatibility haplotype that was fixed in *C. orientalis* was also found in *C. bursa-pastoris*, suggesting that *C. orientalis* became self-compatible before the formation of *C. bursa-pastoris* (Bachmann *et al.*, 2019 [↗](#)); On the other hand, restoring the great polymorphism of functional self-incompatibility haplotypes from a self-compatible ancestral population is very unlikely, therefore self-incompatibility should be the ancestral state of the (*C. grandiflora* + *C. rubella*) lineage. Although we cannot exclude the alternative hypothesis that the progenitor of Cbp_cg is a self-fertilizing ghost population, the most parsimonious hypothesis is that Cbp_cg subgenome originated from outcrossing individuals. For all these reasons, the resynthesized *Capsella* allotetraploids may still provide a realistic approximation to the early stages of natural *C. bursa-pastoris*.

Another limitation of using resynthesized allotetraploids is that we could not completely exclude the effect of colchicine treatment, even though we used second-generation allotetraploids (Münzbergová, 2017 [↗](#)). Colchicine treatment could affect pollen and seed quality and the rate of homoeologous synapsis in resynthesized allotetraploids. Spontaneous *Capsella* allotetraploids have been repeatedly found among diploid hybrids that were not treated with colchicine solution (Bachmann *et al.*, 2021 [↗](#), and own unpublished results). For future studies, these spontaneous allotetraploids would be excellent materials for accurately estimating the rate of homoeologous synapsis in newly formed *Capsella* allotetraploids. Nevertheless, the reported influence of colchicine treatments on the second generation of synthetic polyploids was either trivial (Husband *et al.*, 2016 [↗](#)) or not in the same direction as our results (Münzbergová, 2017 [↗](#)). Hence, the observed pollen and seed quality reduction and rampant homoeologous synapsis were unlikely pure artifacts from colchicine treatment.

Resynthesized and natural *Capsella* allotetraploids had distinct phenotypes

The most noticeable morphological difference between resynthesized and natural *Capsella* allotetraploids was related to the selfing syndrome. Compared to second-generation resynthesized allotetraploids, natural *C. bursa-pastoris* had smaller floral organs, more pollen per flower, and a shorter stem length (Figure 1 [↗](#)). In particular, trait values of petal size and stamen length of the resynthesized allotetraploids had almost no overlap with natural *C. bursa-pastoris* but were similar to the outcrossing progenitor *C. grandiflora*. The shorter stem length in natural *C. bursa-pastoris* may reflect a shorter lifespan, which is also a feature of self-fertilizing species (Duminil *et al.*, 2009 [↗](#); Lesaffre & Billiard, 2020 [↗](#)). If natural *C. bursa-pastoris* indeed originated from the hybridization between *C. grandiflora*-like outcrossing plants and *C. orientalis*-like self-fertilizing plants, the selfing syndrome in *C. bursa-pastoris* does not reflect the instant dominance effect of the *C. orientalis* alleles, but evolved afterward. A remaining question is whether the genetic basis of the selfing syndrome in *C. bursa-pastoris* is the same as in *C. orientalis*. Did the pre-existing selfing syndrome-related alleles from *C. orientalis* facilitate the evolution of selfing syndrome in *C. bursa-pastoris*? Was the selfing syndrome of *C. bursa-pastoris* established by silencing/replacing the *C. grandiflora* alleles or new regulations on both *C. orientalis* and *C. grandiflora* homoeologs?

Although the selfing syndrome in natural *C. bursa-pastoris* was most likely an adaptation to the change in mating system, these morphological changes may be accelerated by the compensation or adaptation to a polyploid state (Hollister, 2015 [↗](#)). WGD directly increases the size of various types of cells (Beaulieu *et al.*, 2008 [↗](#); Katagiri *et al.*, 2016 [↗](#); Snodgrass *et al.*, 2017 [↗](#); Wilson *et al.*, 2021 [↗](#)) and disturbs the efficiency of multiple physiological processes (Drake *et al.*, 2013 [↗](#); Bombles, 2020 [↗](#)). Compared to newly resynthesized autopolyploids, natural autopolyploid plants often have smaller cell or organ sizes (Maherali *et al.*, 2009 [↗](#); Münzbergová, 2017 [↗](#); Landis *et al.*, 2020 [↗](#)), possibly driven by the demand for optimizing physiological processes or resource allocation (Roddy *et al.*, 2020 [↗](#); Domínguez-Delgado *et al.*, 2021 [↗](#)). In the case of allotetraploid

Capsella, the selection of selfing-syndrome-related traits and the adaptation to a polyploid state (e.g., decreasing the size of cell or organ for physiological efficiency or better energy allocation) may work synergistically and can be difficult to separate.

Apart from the selfing-syndrome-related traits, newly resynthesized *Capsella* allotetraploids had lower proportions of viable pollen (**Figure 1l**) and normal seeds (**Figure 1m**). In contrast, pollen and seed quality in natural *C. bursa-pastoris* were much higher, as good as for the diploid species. The higher pollen and seed quality in natural *C. bursa-pastoris* was possibly achieved by improving meiotic behaviors. Meiotic stabilization is another important aspect of adaptation to an allopolyploid state (Blasio *et al.*, 2022). Newly resynthesized allopolyploids suffer more often than natural allopolyploids, from frequent and severe meiosis abnormalities, which are associated with lower pollen viability and fertility in resynthesized allopolyploids (Szadkowski *et al.*, 2010; Zhang *et al.*, 2013; Henry *et al.*, 2014). The molecular basis of improved meiotic synapsis in natural allopolyploids is not completely clear, but several loci that suppress homoeologous synapsis or recombination are essential for the process (Jenczewski *et al.*, 2003; Nicolas *et al.*, 2009; Greer *et al.*, 2012; Martín *et al.*, 2017). The exact mechanism of meiotic stabilization in natural *C. bursa-pastoris* needs further investigation.

The emergence of non-additive gene expression in allotetraploids was a two-stage process

Non-additive gene expression in allotetraploid *Capsella* was altogether limited and neither a complete relict of short-term genomic interactions nor entirely due to gradual divergence. We found that about 40% of the cases of expression level dominance in natural *C. bursa-pastoris* could already be found in the second generation of resynthesized allotetraploids (relict ELDs). Most of these relict ELDs and their directions were shared by the two resynthesized allotetraploid groups (**Figure 4c**, **Figure 4**—figure supplement 1), suggesting that most relict ELDs were repeatable alterations. On the other hand, about 60% of the cases of expression level dominance were specific to natural *C. bursa-pastoris* (Cbp-specific ELDs, **Figure 4c**, and **Figure 4**—figure supplement 1), revealing the contributions from long-term evolution.

The relict ELDs and Cbp-specific ELDs differed in several features. While the vast majority of relict ELDs were up-regulated (97% in flowers and 88% in leaves), Cbp-specific ELDs had a more balanced number of up- and down-regulated ELDs (61% and 54% were up-regulated in flowers and leaves, respectively), suggesting the short and long-term expression level dominance had different molecular basis. In diploid or polyploid interspecific hybrids, overexpression is often more common than underexpression. The trend has been observed in a wide range of organisms, including *Brassica* (Wu *et al.*, 2018; Li *et al.*, 2020; Wei *et al.*, 2021) and *Raphanobrassica* (Ye *et al.*, 2016), cotton (Yoo *et al.*, 2013), brown algae (Sousa *et al.*, 2019) and copepod (Barreto *et al.*, 2015). Results in *Capsella* further showed that short-term mechanisms mainly caused the excess of up-regulated ELDs. Among the short-term mechanisms, intergenomic interaction of regulatory elements is the most likely candidate for generating the excess of up-regulated ELDs, considering that these up-regulated ELDs were highly shared between the two resynthesized allotetraploid groups, and between resynthesized and natural allotetraploids. A global DNA methylation change may also contribute to the excess of up-regulation in resynthesized allotetraploids, if methylation levels were systematically lower in *Capsella* allotetraploids, as observed in *Mimulus* (Edger *et al.*, 2017). However, methylation change alone fails to explain why the majority of these up-regulated ELDs in resynthesized allotetraploids were retained in natural allotetraploids, especially in leaves (**Figure 4c**, **Figure 4**—figure supplement 1).

Besides, the relict ELDs contained more Cg-ELD than Co-ELD, but Cbp-specific ELDs had more Co-ELDs, especially in flowers (**Figure 4c**, **Figure 4**—figure supplement 1). The increase of Co-ELDs in natural *C. bursa-pastoris* mirrored the morphological difference: The floral organ size of resynthesized allotetraploids was similar to that of *C. grandiflora*, whereas natural *C. bursa-*

pastoris was more similar to *C. orientalis* (Figure 1a–g). However, it is worth noting that the reversed trend of expression level dominance may not be the fundamental genetic basis of selfing syndrome, but reflect the different composition of tissue/cells in RNA samples. Both morphological changes and the direction of expression level dominance could result from upstream regulatory changes.

In addition, although unbalanced homoeolog content (not 2:2) caused by homoeologous synapsis was common in our resynthesized allotetraploids, they were still not the main cause of expression level dominance in resynthesized allotetraploids. If expression level dominance were mainly caused by possessing more than two copies of the EL-dominant homoeolog, we would expect the relative expression from the EL-dominant homoeolog to increase and the EL-recessive homoeolog to decrease in genes with significant expression level dominance. In contrast to this expectation, we found that, on average, the expression of EL-dominant homoeologs (relative to the transcriptome of the subgenome) was similar to that in diploid parental species, while the expression of EL-recessive homoeologs changed toward the EL-dominant homoeolog (Figure 8). This suggests that expression level dominance is mainly achieved by altering the expression of EL-recessive homoeologs. This result is consistent with studies in a wide range of allopolyploid organisms (Yoo *et al.*, 2013; Cox *et al.*, 2014; Combes *et al.*, 2015; Sousa *et al.*, 2019), though not in resynthesized *Brassica napus* (Wu *et al.*, 2018). This conservative pattern can be explained by intergenomic interaction between divergent regulatory elements (Hu & Wendel, 2019), but direct evidence is still lacking.

As for transgressive gene expression, we found almost no TRE genes in resynthesized allotetraploids, but a mere 1.3% TRE genes in natural *C. bursa-pastoris*, with a threshold of $FC > 2$ (Figure 4—Source Data 1). In agreement with several previous observations (Flagel & Wendel, 2010; Yoo *et al.*, 2013; Zhang *et al.*, 2016b), the results in *Capsella* suggest that transcriptional novelties in allopolyploids were not an instant outcome of allopolyploidization but mainly arose during long-term evolution and remained altogether rather limited.

Homoeologous synapsis was common in resynthesized *Capsella* allotetraploids, and may still be a source of variation in natural *Cbp*

Disomic inheritance in allopolyploid species is not established all at once (Henry *et al.*, 2014), and strict disomic inheritance may have never been achieved in some allopolyploid species. In contrast to the disomic inheritance and the low level of homoeolog expression loss in natural *C. bursa-pastoris*, we found abundant traces of homoeologous segregation or recombination in all twelve lines of resynthesized *Capsella* allotetraploids, after only one meiosis. The observation is in line with many recent studies in which abundant homoeologous exchanges were found in allopolyploids (Lloyd *et al.*, 2018; Pelé *et al.*, 2018). The contrast between resynthesized and natural allotetraploids suggested that preferential synapsis was rapidly improved in natural *C. bursa-pastoris*, and/or a balanced number of homoeologs was strongly preferred by selection, otherwise, we would expect a much higher proportion of homoeolog replacement (having four copies of the same homoeolog) after 100,000-year recurrent self-fertilization with tetrasomic/heterosomic inheritance.

Homoeologous synapsis was the major mechanism for the variation of homoeolog expression bias and homoeolog expression loss in resynthesized *Capsella* allopolyploids (Figure 5,7). Several models have been proposed to explain homoeolog expression bias and biased genome fractionation in allopolyploids, including different TE contents (Woodhouse *et al.*, 2014; Cheng *et al.*, 2016; Wendel *et al.*, 2018), the epigenetic difference of subgenomes (Li *et al.*, 2014), different strength of regulatory elements, as a result of enhancer runaway (Fyon *et al.*, 2015; Bottani *et al.*, 2018). It was also suggested that the initial homoeolog expression bias may be reinforced in long-term evolution, as the homoeolog with a lower initial expression level is subject

to weaker purifying selection, and has a larger chance of degeneration (Woodhouse *et al.*, 2014). However, in resynthesized *Capsella* allotetraploids, homoeologous synapsis played an important role in generating homoeolog expression bias variation, possibly overshadowing the influence of other mechanisms. This result does not conflict with the observed association between TE content in parental species and genome dominance (Woodhouse *et al.*, 2014). While TE contents may have only a minor effect in directly generating homoeolog expression bias variation, they could still be informative in predicting homoeolog expression bias in established allopolyploids, as the presence of TEs in the flanking regions may affect the fitness effect of homoeolog expression bias variation (Hollister & Gaut, 2009). In other words, TEs may not function as a strong mutagenic mechanism of homoeolog expression bias variation, but affect the selection on homoeolog expression bias variation, as a form of genetic load.

For established natural allotetraploids, occasional homoeologous synapsis may still be an important source of genetic variation, even long after the allopolyploidization event. Although an earlier allozymic study (Hurka *et al.*, 1989) and an approximate Bayesian computation (ABC) with high throughput sequencing data (Roux & Pannell, 2015) suggest that natural *C. bursa-pastoris* exhibits disomic inheritance, neither analysis could reject homoeologous synapsis at a lower rate. A very small proportion of homoeologous synapsis may be negligible for inferring the dominant mode of inheritance, but in terms of causing homoeologous gene loss and creating genetic variation, homoeologous synapsis can still be more influential than point mutations. Due to the inevitable technical variation of RNA-seq and expression variation across genes, we were unable to confidently resolve smaller blocks of unbalanced homoeolog content. Despite the small sample size and the low resolution of RNA-seq data, we noticed that some small genomic blocks with homoeolog replacement were shared by the individuals of the same population but varied among populations of natural *C. bursa-pastoris* (Figure 6, Figure 6—figure supplement 1 and 2), suggesting that homoeologous synapsis still contribute to expressional variation in natural *C. bursa-pastoris*. Apart from homoeolog synapsis in a single-origin allopolyploid species, unbalanced content of homoeologs could also arise from secondary introgression from diploid parental species. Detailed demographic modelling would be needed for distinguishing the two scenarios.

Conclusion

In conclusion, together with Duan *et al.* (2023), the present study shows that both short- and long-term mechanisms contributed to transcriptomic and phenotypic changes in natural allotetraploids. However, the initial gene expression changes were largely reshaped during long-term evolution leading to more pronounced morphological changes. Resource limitations and/or adaptation to self-fertilization also, drive flowers' evolution after polyploidization.

Material and Methods

Plant material

The present study used five *Capsella* plant groups (Figure 1, Figure 1—Source Data 1), including diploid *C. orientalis* (Co2), diploid *C. grandiflora* (Cg2), two types of resynthesized allotetraploids (Sd and Sh), and natural allotetraploids, *C. bursa-pastoris* (Cbp). RNA-sequencing data and most phenotypic data (except floral morphologic traits) of Co2, Cg2, Sd, and Sh groups were from Duan *et al.* (2023). The Sd and Sh allotetraploids only differed in the order of hybridization and whole genome duplication. Allotetraploids of the Sd group were generated by crossing synthetic autotetraploid *C. orientalis* with autotetraploid *C. grandiflora*, whereas the Sh group was generated by inducing WGD in the first generation of diploid hybrids of *C. orientalis* × *C.*

grandiflora. In all interspecific crosses, *C. orientalis* served as maternal plants, mimicking the formation of the natural allotetraploid species, *C. bursa-pastoris* (Hurka *et al.*, 2012). All *C. orientalis* plants used in the experiment are descendants of one wild *C. orientalis* individual, and all the *C. grandiflora* plants are descendants of three individuals from one *C. grandiflora* population (Figure 1c and Figure 1–Source Data 1).

To compare the resynthesized allotetraploids with natural allotetraploids, natural *C. bursa-pastoris* was added to the present study. Seeds of wild *C. bursa-pastoris* plants were grown in a growth chamber for one generation. Then the second generation of *C. bursa-pastoris* plants was grown in the same experiment together with the other four plant groups used in Duan *et al.* (2023). All five plant groups were grown in a growth chamber under long-day conditions (16-h light at 22°C and 8-h dark at 20°C, light intensity=137 uE-m⁻²-s⁻¹).

Each of the five plant groups was represented by six “lines”, and each line had six individuals, which were full siblings from either self-fertilization (Co2, Cbp, Sh, and Sd groups) or brother-sister mating (Cg2 group). The six lines of *C. bursa-pastoris* were from six populations (Figure 1c and Figure 1–Source Data 1), representing three major genetic clusters of the wild *C. bursa-pastoris* (Kryvokhyzha *et al.*, 2019b). Each line originated from an independent allopolyploidization event for the Sh and Sd groups. For Co2 and Cg2 groups, “line” only referred to the offspring of one parental plant (Co2) or a pair of parental plants (Cg2) in the previous generation.

Plants used in the present study and a previously published work (Duan *et al.*, 2023) were different subsets of a single experiment. The entire experiment had eight plant groups, including the five plant groups used in the present study (Groups Cg2, Co2, Sh, Sd, and Cbp; 5 groups x 6 lines x 6 individuals = 180 plants), and other three groups that were only used in Duan *et al.* (2023), Groups F, Co4 and Cg4; 3 groups x 6 lines x 6 individuals = 108 plants). All these plants were grown in 36 trays placed on six shelves inside the same growth chamber. Each tray had exactly one plant from each of the eight groups, and the positions of the eight plants within each tray (A-H) were randomized with random.shuffle() method in Python (Supplementary file 1). The position of the 36 trays inside the growth room (1-36) was also random and the positions of all trays were shuffled once again 28 days after germination (randomized with RAND() and sorting in Microsoft Excel Spreadsheet).

Phenotyping

Floral morphological traits were measured for all five groups on 165 plants 7-14 days after the first flower opened. The rest 15 plants were not measured due to technical errors. Two fully opened young flowers were dissected for each plant, and the floral organs were scanned with a photo scanner (Epson Perfection V370) at 3200 dpi. Seven floral morphological traits were measured on the digital images with Fiji, an open-source platform for biological-image analysis (Schindelin *et al.*, 2012), including sepal width, sepal length, petal width, petal length, pistil width, pistil length, and stamen length. For each plant, two flowers were examined. Three sepals, petals and stamens, and one pistil were measured for each flower.

Stem length (total sample size n=171), flowering time (n=170), pollen traits (n=142), and seed traits (n=151) were measured for the five plant groups. Measurements of the Cg2, Co2, Sh, and Sd groups were from Duan *et al.* (2023), and measurements of the Cbp group were added by the present study, which were obtained in the same way as the other four groups. The length of the longest stem (stem length) was measured on dry plants. The number of days from germination start to the opening of the first flower was recorded as flowering time. The number of pollen grains per flower (pollen counts) was calculated by counting 1/60 (Co2, Sd, Sh, and Cbp groups) or 1/120 (Cg2 group) of the total pollen grains of a flower using a hemocytometer. Pollen viability was measured with an aceto-carmin staining method (Duan *et al.*, 2023), by examining at least 300 pollen grains per flower. Pollen counts and pollen viability were measured on two flowers of each

individual. Seeds from the 11th to 20th fruits were counted and were used to measure the proportion of normal seeds. In the case of the Cg2 and Cg4 groups, seeds were obtained through hand pollination. The first ten fruits were used when not all of these flowers set fruits. Individuals with less than ten fruits were excluded from the analysis. Seeds that were flat or dark and small were considered abnormal.

General linear models (for non-ratio data) or generalized linear models with quasi-binomial error distribution and a logit link function (for pollen viability and seed quality) were fit to each phenotypic trait. The difference in phenotypic traits among the five plant groups was tested with one-way analyses of variance (ANOVA, trait value~ plant group, type-III tests), using R package “car” version 3.1-2 (Fox & Weisberg, 2019 [DOI](#)). A two-way ANOVA (trait value~ plant group + tray ID, type-III tests) was also tried to test the positional effect. When plant group had a significant effect on a trait, groups with different means were identified by Tukey’s HSD test using R packages “agricolae” version 1.3-6 (de Mendiburu & Yaseen, 2020 [DOI](#)) and “multcomp” version 1.4-25 (Hothorn *et al.*, 2008 [DOI](#)).

RNA-sequencing

RNA-sequencing was conducted for the five plant groups (Co2, Cg2, Sd, Sh, and Cbp). For each line, leaf and young inflorescence (flower) samples from one randomly chosen individual were sequenced, resulting in 60 RNA-sequencing samples (5 groups × 6 lines × 2 tissues). RNA-sequencing data of the Co2, Cg2, Sd, and Sh groups were from Duan *et al.* (2023) [DOI](#). Data from the Cbp group was added to the present study, but the Cbp samples were collected and sequenced simultaneously with the other four plant groups in 2019. The 8th and 9th leaves were harvested at the emergence of the 11th leaf, and 2-4 inflorescences with only unopened flower buds were collected 7-14 days after the first flower opened. The collected tissue was frozen in liquid nitrogen and stored at -80 °C.

Total RNA was extracted from leaf and flower samples with a cetyl-trimethyl-ammonium-bromide (CTAB) based method (Duan *et al.*, 2023 [DOI](#)). DNA contamination was further removed by the RNase-Free DNase Set (QIAGEN). RNA libraries were prepared with Illumina TruSeq Stranded mRNA (poly-A selection) kits and sequenced with pair-end reads of 150 bp on three NovaSeq 6000 S4 lanes, by the SNP&SEQ Technology Platform in Uppsala, Sweden. One sequencing library was generated for each diploid sample, and two libraries were generated for each allotetraploid sample.

Gene and homoeolog expression

Raw RNA-seq reads were mapped to the reference genome of *Capsella rubella* (Slotte *et al.*, 2013 [DOI](#)) using Stampy v.1.0.32 (Lunter & Goodson, 2011 [DOI](#)). The expected divergence between reference and query sequences was set to 0.02, 0.04, and 0.025 for *C. grandiflora*, *C. orientalis*, and allotetraploids, respectively. Mapping quality was inspected by Qualimap v. 2.2.1 (Okonechnikov *et al.*, 2016 [DOI](#)). The number of reads mapped to each gene was counted by HTSeq v.0.12.4 (Anders *et al.*, 2015 [DOI](#)), using the mode “union” (hereinafter referred to as “unphased gene expression”). The average number of mapped reads was 38.4±2.4 and 70.0±3.5 for the diploid and tetraploid samples, respectively (Supplementary file 2). For all analyses on unphased gene expression, the mapped reads were downsampled with a custom Python script (Duan *et al.*, 2023 [DOI](#)), so that all five groups had a similar average number of mapped reads.

The expression level of separate homoeologs in allotetraploids was determined by the program HyLiTE v.2.0.2 (Duchemin *et al.*, 2015 [DOI](#)). Alignment results from the software Stampy v.1.0.32 (Lunter & Goodson, 2011 [DOI](#)) of all five groups and the *C. rubella* reference genome (Slotte *et al.*, 2013 [DOI](#)) were used as the input for HyLiTE. HyLiTE performed SNP calling, classified RNA-seq

reads of allotetraploids to parental types according to the identified diagnostic variation between the two diploid parental species and generated a table of homoeolog read counts for allotetraploid individuals (hereinafter referred to as “phased gene expression”).

After partitioning the homoeolog expression, the average library size of allotetraploid subgenomes was similar to the library size of diploid groups (one-way ANOVA, $F_{4,91}=1.28$, $p = 0.29$). To reduce bias between phased and unphased expression datasets, when the homoeolog expression of allotetraploids was compared with gene expression in diploid parental species, the expression level of each gene in diploid individuals was rescaled by the proportion of reads that can be phased for the same gene in allotetraploid individuals.

The overall gene expression pattern of five plant groups in each tissue was visualized by MDS analysis, using the R package “edgeR” (version 3.28.1; Robinson et al., 2010) in the R software environment version 3.6.3 (R Core Team, 2020). For both phased and unphased gene expression data, genes with count-per-million (CPM) over one in at least two samples were used for the MDS analysis, and expression levels were normalized with the trimmed mean of M-values (TMM) method.

Differential expression (DE) analysis

DE analysis was conducted on both unphased and phased gene expression data with the R package “edgeR” (version 3.28.1; Robinson et al., 2010), using TMM normalized gene expression levels. A negative binomial generalized linear model (GLM) was fitted to each dataset. Pairwise group contrasts were then made for each GLM model, and gene-wise quasi-likelihood F-tests were conducted to detect expression changes in each contrast. For unphased data, pairwise contrasts were made among the original five groups (Co2, Cg2, Sd, Sh, and Cbp). For phased data, allopolyploid subgenomes were treated as separate groups (Sd_co, Sd_cg, Sh_co, Sh_cg, Cbp_co, and Cbp_cg) and were compared with the two diploid groups (Co2, Cg2). Genes with an expression fold-change (FC) larger than two and a false discovery rate (FDR) less than 0.05 were considered significant differentially expressed genes (DEGs).

Expression level dominance and transgressive expression

To measure the extent of non-additive expression in allotetraploids, genes were classified into ten expression categories (Table 1), by comparing the total expression of both homoeologs in an allotetraploid group to the gene expression level in a diploid group. The ten categories were modified from the classification by Zhang et al. (2016). Results of DE analysis on unphased gene expression ($FC > 2$ and $FDR < 0.05$) were used for the classification.

Homoeolog expression bias

Homoeolog expression bias of gene i in allotetraploid individual j was measured by the proportion of cg-homoeolog expression (cg_{ij}) in the total expression of both homoeologs ($cg_{ij}/(cg_{ij}+co_{ij})$). The distribution of homoeolog expression bias was then viewed by individuals, chromosomes, or along genomic coordinates. Signs of the segregation or recombination of homoeologous chromosomes were revealed in resynthesized allopolyploids by the distribution of homoeolog expression bias along genomic coordinates.

The copy number of cg- and co-homoeologs of each gene and the breakpoints between chromosomal segments resulting from homoeologous recombination were estimated with a five-state Hidden Markov Model (HMM) of gene-wise homoeolog expression bias, using a modified version of the R package “HMMcopy” version 1.40.0 (Lai et al., 2022). The five states corresponded to (0, 1, 2, 3, 4) gene copies from Cg and (4, 3, 2, 1, 0) copies from Co, respectively. The expected optimal values of median homoeolog expression bias (m) were set to 0.01, 0.25, 0.5, 0.75, and 0.99 for the five states. The length of segments was controlled by arguments e and $strength$. e

Group	Description	Classification criteria
a	Additive expression with no parental differentiation	$Cg_i = x_{ij} = Co_i$
b	Partial ELD or additive expression with parental differentiation	$(Cg_i < x_{ij} < Co_i)$ or $(Co_i < x_{ij} < Cg_i)$ or $(Cg_i \neq Co_i \text{ and } x_{ij} = Cg_i \text{ and } x_{ij} = Co_i)$
c	Up-regulated ELD toward Cg2	$x_{ij} = Cg_i \text{ and } x_{ij} > Co_i$
d	Down-regulated ELD toward Cg2	$x_{ij} = Cg_i \text{ and } x_{ij} < Co_i$
e	Up-regulated ELD toward Co2	$x_{ij} = Co_i \text{ and } x_{ij} > Cg_i$
f	Down-regulated ELD toward Co2	$x_{ij} = Co_i \text{ and } x_{ij} < Cg_i$
g	Up-regulated TRE with no parental differentiation	$Cg_i = Co_i \text{ and } x_{ij} > Cg_i \text{ and } x_{ij} > Co_i$
h	Up-regulated TRE with parental differentiation	$Cg_i \neq Co_i \text{ and } x_{ij} > Cg_i \text{ and } x_{ij} > Co_i$
i	Down-regulated TRE with no parental differentiation	$Cg_i = Co_i \text{ and } x_{ij} < Cg_i \text{ and } x_{ij} < Co_i$
j	Down-regulated TRE with parental differentiation	$Cg_i \neq Co_i \text{ and } x_{ij} < Cg_i \text{ and } x_{ij} < Co_i$

* Cg_i : expression level of gene i in the Cg2 group; Co_i : expression level of gene i in the Co2 group; x_{ij} : expression level of gene i in allotetraploid group j, and $j \in (Sd, Sh, Cbp)$; ELD: expression level dominance; TRE: transgressive expression; The significance of differential expression between groups were determined by results of differential expression analysis on unphased gene expression, with a threshold of fold-change > 2 and false discovery rate < 0.05 .

Table 1

Classification of additive and non-additive gene expression pattern in allotetraploids

is the initial value of the transition probability that the state (copy number of cg homoeolog) does not change between two adjacent genes, and *strength* is the strength of this initial *e*. Smaller values of *strength* increase the flexibility of transition probability, and an extremely large value leads to almost fixed transition probabilities. For natural *C. bursa-pastoris*, *e* and *strength* were set to $(1 - 1e-7)$ and $1e+7$, respectively. For resynthesized allotetraploids, *e* was increased to $(1 - 1e-10)$, and *strength* was increased to $1e+12$, as homoeologous exchanges were expected to be rare for resynthesized allotetraploids which had only experienced one round of meiosis.

The effect of homoeologous segregation and recombination on homoeolog expression bias in resynthesized allotetraploids was analyzed by a generalized linear model with quasi-binomial error distribution and logit link function. Gene-wise homoeolog expression bias was reshaped as binomial data (read counts from cg homeolog and total counts from both homoeologs). The copy number of cg homoeologs (0, 1, 2, 3, and 4) estimated by HMM segmentation was used as the explanatory variable. The variance of homoeolog expression bias explained by the estimated copy number of cg homoeolog was assessed by the coefficient of determination (R^2), calculated with R package “rsq” version 2.5 (Zhang, 2022 [↗](#)).

The relationship between non-additive gene expression and homoeolog expression was explored by comparing the estimated expression fold change of each homoeolog among additive and different non-additive expression categories. The expression fold change of homoeologs was measured by homoeolog expression of gene *i* in allotetraploid group *k* divided by expression of gene *i* in the corresponding diploid group, i.e., $\log_2FC(cg_{ik}/Cg2_i)$ or $\log_2FC(co_{ik}/Co2_i)$, using the estimated FC from DE analysis on phased data. The difference of expression fold change between EL-dominant and EL-recessive homoeologs was tested by Welch’s two-sample t-tests.

Loss of homoeolog expression

The most extreme homoeolog expression bias occurs when one homoeolog is silenced or lost in allotetraploids. To explore the timing and mechanism of the loss of homoeolog expression in *Capsella*, the number of genes with homoeolog expression loss was counted for each resynthesized or natural allotetraploid. Lowly or occasionally expressed genes were excluded from the analysis to reduce noise from sequencing and phasing. Specifically, if one homoeologous gene had obvious expression ($CPM > 5$) in all six individuals of the corresponding diploid species, but had almost no expression in one allotetraploid individual ($CPM < 0.5$), the case was considered a loss of homoeologous expression.

Gene ontology (GO) enrichment analysis

GO enrichment analysis was performed with R package “topGO” version 2.52.0 (Alexa & Maintainer, 2023 [↗](#)), using results of DE analysis on unphased gene expression and GO annotations of *C. rubella* downloaded from PlantRegMap (Tian *et al.*, 2020 [↗](#)). Genes that were differentially expressed in both Cbp-Sd and Cbp-Sh comparisons ($FC > 1.5$ and $FDR < 0.05$; 578 genes in flowers and 345 in leaves) were used as the test set, and all genes with GO annotation and $CPM > 1$ in at least two samples were used as the background set (17 718 genes in flowers and 15 631 genes in leaves). GO terms were scored with the “classic” algorithm and Fisher’s exact tests, and p-values were adjusted with the Benjamini-Hochberg procedure in R. GO terms with less than 10 annotated genes were excluded from the analysis.

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Data availability

The RNA-sequencing data of natural *C. bursa-pastoris* generated by this article are available in NCBI Sequence Read Archive (SRA) and can be accessed with BioProject number PRJNA848625. BioSample accessions are listed in Supplementary file 2. Unphased and phased expression datasets, phenotypic measurements, and R scripts used for statistical tests and visualization are available on GitHub (https://github.com/disporum/RNA104_paper2 )

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Editors

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Reviewer #1 (Public Review):

This study aims to determine the relative importance of the immediate effects of allopolyploidization from subsequent evolution in phenotypic traits associated with the selfing syndrome and in gene expression traits in the selfing allopolyploid *Capsella bursa-pastoris* and its diploid progenitors *Capsella grandiflora*, which is outcrossing, and *Capsella orientalis*, which is selfing. To do this, they compared five categories of plant: the two progenitors of the allopolyploid, hybrids resynthesized from the progenitors with a whole-genome duplication either before or after the hybridization event, and the naturally occurring allopolyploid.

Two lines of evidence were used: phenotypic data from the plants grown in a common environment, and RNAseq data from a subset of the plants.

The phenotypic data indicate that the selfing syndrome of *C. bursa-pastoris* likely evolved after the initial allopolyploidization event, and that pollen and seed viability recovered following the allopolyploidization event. The results are compelling but would benefit from small clarifications to the methods and statistics to account for possible positional effects in the growth chamber. Using a linear mixed model rather than a simple ANOVA would solve this problem.

The RNAseq data are used to explore overall expression patterns (using multi-dimensional scaling), patterns of differential expression (additive, dominant, or transgressive), and homeolog expression bias, and to determine the relative contributions of the original allopolyploidization event and subsequent evolution. Statistical cutoffs were used to categorize gene expression patterns, but the description and categorization of these patterns appears to have been largely qualitative, and might be strengthened by including more statistical detail in questions like whether homeologous expression bias did indeed show more variation in resynthesized and evolved allopolyploids.

The study includes evidence that homeolog expression bias (overrepresentation of an allele from one species) results in part from homeologous synapsis (uneven inheritance of chromosome segments). These deviations from patterns consistent with 2:2 inheritance of genomic regions are highly variable between individuals in resynthesized allopolyploids but appeared to be mostly consistent within (but not between) populations in natural *C. bursa-pastoris*. This is intriguing evidence that segregation can be an important source of variation in allopolyploids. However, it was limited by the difficulty of inferring homeologous recombination breakpoints with RNAseq data because of the scale of recombination in wild populations (rather than resynthesized allopolyploids). In future identifying such breakpoints will be an interesting direction for this and other allopolyploid systems.

This research suggests many follow-up questions. In particular, it may be possible to identify evidence about the mechanism of the original hybridization event. How frequently do unreduced gametes occur in these species, and is it likely that *C. bursa-pastoris* evolved via a triploid bridge? Exploring the viability, fertility, and phenotypes of triploids produced in both directions could be a valuable future direction.

Future research, or the current study, could also valuably explore what kinds of genes experienced what forms of expression evolution. A brief description of GO terms frequently represented in genes which showed strong patterns of expression evolution might be suggestive of which selective pressures led to the changes in expression in the *C. bursa-pastoris* lineage, and to what extent they related to adaptation to polyploidization (e.g. cell-cycle regulators), compensating for the initial pollen and seed inviability or adapting to selfing (endosperm- or pollen-specific genes), or adaptation to abiotic conditions.

Overall, this is an interesting and valuable contribution to the field's understanding of how expression evolves in interaction with hybridization and polyploidy. Particularly in combination with the team's previous study on these lines, this experimental design is effective for separating the contributions of hybridization, WGD, and evolution over time.

Update: the authors have thoughtfully and thoroughly updated the manuscript to address all the questions I raised. I appreciate the chance to review this valuable contribution to the scientific literature.

- <https://doi.org/10.7554/eLife.88398.2.sa1>

Reviewer #2 (Public Review):

The flowering plant *Capsella bursa-pastoris* is an allotetraploid formed from the genomes of *Capsella orientalis* and *Capsella grandiflora*. An outstanding question in the evolution of allotetraploids is the relative contribution of immediate consequences of allopolyploidization vs. long-term evolution after the event. The authors address this question by re-synthesizing the allotetraploid in the lab using the two progenitor species, and comparing its phenotypic and gene expression variation to naturally occurring *C. bursa-pastoris*. They find evidence primarily for long-term phenotypic evolution towards a selfing syndrome in *C. bursa-pastoris*, and a combination of short and long-term changes to gene expression.

The manuscript is thorough and provides lots of new insights into the mechanisms driving evolution in allopolyploids. I especially appreciated the detailed examination of different mechanisms driving gene expression variation. There are some important limitations of the experimental design related to independent evolution of the progenitor species and effects of the colchicine treatment used to induce polyploidy, but these are well-addressed in the Discussion.

- <https://doi.org/10.7554/eLife.88398.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Public Reviews:

*The study could also valuably explore what kinds of genes experienced what forms of expression evolution. A brief description of GO terms frequently represented in genes which showed strong patterns of expression evolution might be suggestive of which selective pressures led to the changes in expression in the *C. bursa-pastoris* lineage, and to what extent they related to adaptation to polyploidization (e.g. cell-cycle regulators), compensating for the initial pollen and seed inviability or adapting to selfing (endosperm- or pollen-specific genes), or adaptation to abiotic conditions. "*

We did not include a gene ontology (GO) analysis in the first place as we did not have a clear expectation on the GO terms that would be enriched in the genes that are differentially expressed between resynthesized and natural allotetraploids. Even if we only consider adaptive changes, the modifications could occur in various aspects, such as stabilizing meiosis, adapting to the new cell size, reducing hybrid incompatibility and adapting to self-fertilization. And each of these modifications involves numerous biological processes and molecular functions. As we could make post-hoc stories for too many GO terms, extrapolating at this stage have limited implications and could be misleading.

Nonetheless, we are not the only study that compared newly resynthesized and established allopolyploids. GO terms that were repeatedly revealed by this type of exploratory analysis may give a hint for future studies. For this reason, now we have reported the results of a simple GO analysis.

Recommendations for the authors: please note that you control which, if any, revisions, to undertake

The majority of concerns from reviewers and the reviewing editor are in regards to the presentation of the manuscript; that the framing of the manuscript does not help the general reader understand how this work advances our knowledge of allopolyploid evolution in the broad sense. The manuscript may be challenging to read for those who

aren't familiar with the study system or the genetic basis of polyploidy/gene expression regulation. Further, it is difficult to understand from the introduction how this work is novel compared to the recently published work from Duan et al and compared to other systems. Because eLife is a journal that caters to a broad readership, re-writing the introduction to bring home the novelty for the reader will be key.

Additionally, the writing is quite technical and contains many short-hands and acronyms that can be difficult to keep straight. Revising the full text for clarity (and additionally not using acronyms) would help highlight the findings for a larger audience.

Reviewer #1 (Recommendations For The Authors):

Most of my suggestions on this interesting and well-written study are minor changes to clarify the writing and the statistical approaches.

The use of abbreviations throughout for both transcriptional phenomena and lines is logical because of word limits, but for me as a reader, it really added to the cognitive burden. Even though writing out "homoeolog expression bias" or "hybridization-first" every time would add length, I would find it easier to follow and suspect others would too.

Thank you for this suggestion. Indeed, using less uncommon acronyms or short-hands should increase the readability of the text for broader audience. Now in most places, we refer to "Sd/Sh" and "Cbp" as "resynthesized allotetraploids" and "natural allotetraploids", respectively. We have also replaced the most occurrences of the acronyms for transcriptional phenomena (ELD, HEB and TRE) with full phrases, unless there are extra attributes before them (such as "Cg-/Co-ELD" and "relic/Cbp-specific ELD").

It would be helpful to include complete sample sizes to either a slightly modified Figure 1 or the beginning of the methods, just to reduce mental arithmetic ("Each of the five groups was represented by six "lines", and each line had six individuals" so there were 180 total plants, of which 167 were phenotyped - presumably the other 13 died? - and 30 were sequenced).

The number 167 only applied to floral morphological traits ("Floral morphological traits were measured for all five groups on 167 plants..."), but the exact total sample size for other traits differed. Now the total sample sizes of other traits have also been added to beginning of the second paragraph of the methods.

For this study 180 seedlings have been transplanted from Petri dishes to soil, but 8 seedlings died right after transplanting, seemingly caused by mechanical damage and insufficient moistening. Later phenotyping (2020.02-2020.05) was also disrupted by the COVID-19 pandemic, and some individuals were not measured as we missed the right life stages. Specifically, 5 individuals were missing for floral morphological traits (sepal width, sepal length, petal width, petal length, pistil width, pistil length, and stamen length), 30 for pollen traits, 1 for stem length, and 2 for flowering time. As for seed traits, we only measured individuals with more than ten fruits, so apart from the reasons mentioned above, individuals that were self-incompatible and had insufficient hand-pollination were also excluded. We spotted another mistake during the revision: two individuals with floral morphological measurements had no positional information (tray ID). These measurements were likely mis-sampled or mislabeled, and were therefore excluded from analysis. We assumed most of these missing values resulted from random technical mistakes and were not directly related to the measured traits.

In general, the methods did a thorough job of describing the genomics approaches but could have used more detail for the plant growth (were plants randomized in the growth

chamber, can you rule out block/position effects) and basic statistics (what statistical software was used to perform which tests comparing groups in each section, after the categories were identified).

When describing the methods, mention whether the plants; this should be straightforward as a linear model with position as a covariate.

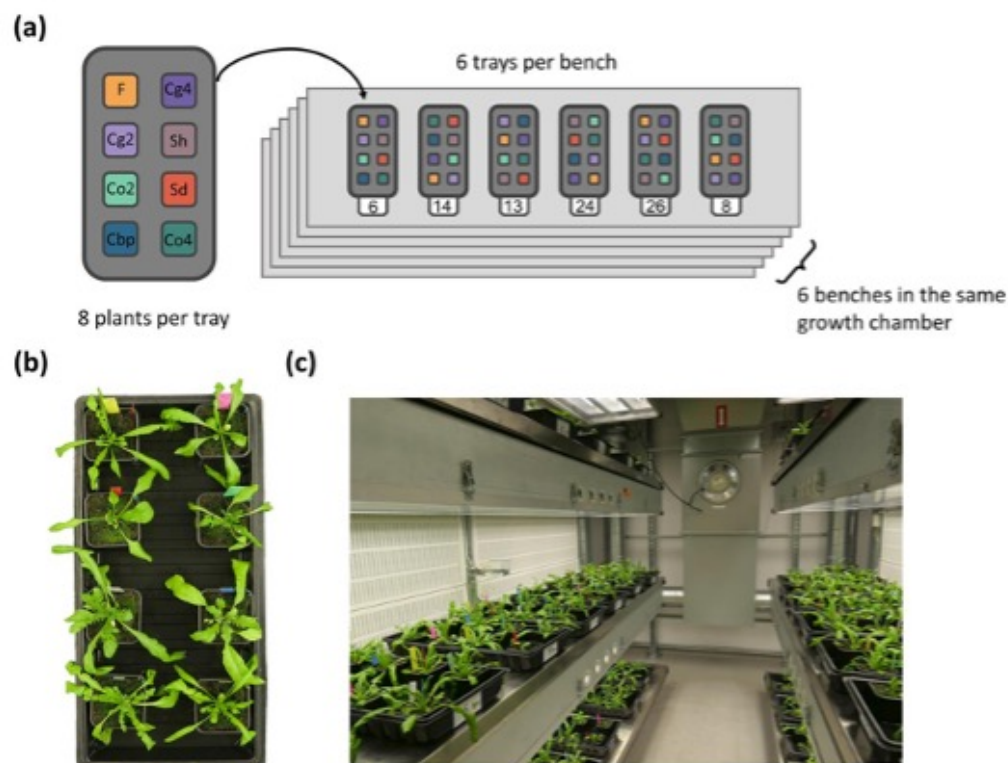
Data used in the present study and a previously published work (Duan et al., 2023) were different subsets of a single experiment. For this reason, we spent fewer words in describing shared methods in this manuscript but tried to summarize some methods that were essential for understanding the current paper. But as you have pointed out, we did miss many important details that should have been kept. Now we have added some description and a table (Supplementary file 1) in the “Plant material” section for explaining randomization, and added more information of the software used for performing statistic tests in the “Phenotyping” section.

Although we did not mention in the present manuscript, we used a randomized block design for the experiment (Author response image 1).

Author response image 1.

Plant positions inside the growth chamber.

Plants used in the present study and Duan et al. (2023) were different subsets of a single experiment. The entire experiment had eight plant groups, including the five plant groups used in the present study (diploid *C. orientalis* (Co2), diploid *C. grandiflora* (Cg2), “whole-genome-duplication-first” (Sd) and “hybridization-first”(Sh) resynthesized allotetraploids, and natural allotetraploids, *C. bursa pastoris* (Cbp), as well as three plant groups that were only used in Duan et al. (2023; tetraploid *C. orientalis* (Co4), tetraploid *C. grandiflora* (Cg4) and diploid hybrids (F)). Each of the eight plant groups had six lines and each line represented by six plants, resulting in 288 plants (8 groups x 6 lines x 6 individuals = 288 plants). The 288 plants were grown in 36 trays placed on six shelves inside the same growth chamber. Each tray had exactly one plant from each of the eight groups, and the position of the eight plants within each tray (A-H) were randomized with `random.shuffle()` method in Python (Supplementary file 1). The position of the 36 trays inside the growth room (1-36) was also random and the positions of all trays were shuffled once again 28 days after germination (randomized with RANDO) and sorting in Microsoft Excel Spreadsheet). (a) Plant distribution; (b) An example of one tray; (c) A view inside the growth chamber, showing the six benches.



With the randomized block design and one round of shuffling, positional effect is very unlikely to bias the comparison among the five plant groups. The main risk of not adding positions to the statistical model is increasing error variance and decreasing the statistical power for detecting group effect. As we had already observed significant among-group variation in all phenotypic traits (p -value $< 2.2e-16$ for group effect in most tests), further increasing statistical power is not our primary concern. In addition, during the experiment we did not notice obvious difference in plant growth related to positions. Although we could have added more variables to account for potential positional effects (tray ID, shelf ID, positions in a tray etc.), adding variables with little effect may reduce statistical power due to the loss of degree of freedom.

Due to one round of random shuffling, positions cannot be easily added as a single continuous variable. Now we have redone all the statistical tests on phenotypic traits and included tray ID as a categorical factor (Figure 2-Source Data 1). In general, the results were similar to the models without tray ID. The F-values of group effect was only slightly changed, and p-values were almost unchanged in most cases (still $< 2.2e-16$). The tray effect ($df=35$) was not significant in most tests and was only significant in petal length (p -value=0.0111), sepal length (p -value=0.0242) and the number of seeds in ten fruits (p -value=0.0367). As expected, positions (tray ID) had limited effect on phenotypic traits.

Figure 2 - I assume the numbers at the top indicate sample sizes but perhaps add this to the figure caption.

Statistical power depends on both the total sample size and the sample size of each group, especially the group with the fewest observations. We lost different number of measurements in each phenotypic trait, and for pollen traits we did have a notable loss, so we chose to show sample sizes above each group to increase transparency. Since we had five different sets of

sample sizes (for floral morphological traits, stem length, days to flowering, pollen traits and seed traits, respectively), it would be cumbersome to introduce all 25 numbers in figure caption and could be hard for readers to match the sample sizes with results. For this reason, we would like to keep the sample sizes in the figure, and now we have modified the legend to clarify that the numbers above groups are sample sizes.

The trend has been observed in a wide range of organisms, including ...' - perhaps group Brassica and Raphanobrassica into one clause in the sentence, since separating them out undermines the diversity somewhat.

Indeed, it is very strange to put “cotton” between two representatives from Brassicaceae. Now the sentence is changed to “... including Brassica (Wu et al., 2018; Li et al., 2020; Wei et al., 2021) and Raphanobrassica (Ye et al., 2016), cotton (Yoo et al., 2013)...”

The diagrams under the graph in Figure 4B are particularly helpful for understanding the expression patterns under consideration! I appreciated them a lot!

Thank you for the comment. We also feel the direction of expression level dominance is convoluted and hard to remember, so we adopted the convention of showing the directions with diagrams.

Reviewer #2 (Recommendations For The Authors):

The science is very interesting and thorough, so my comments are mostly meant to improve the clarity of the manuscript text:

- I found it challenging to remember the acronyms for the different gene expression phenomena and had to consistently cross-reference different parts of the manuscript to remind myself. I think using the full phrase once or twice at the start of a paragraph to remind readers what the acronym stands for could improve readability.*

Thank you for this reasonable suggestion. Now we have replaced the most occurrence of acronyms with the full phrases.

- There are some technical terms, such as "homoeologous synapsis" and "disomic inheritance", which I think are under-defined in the current text.*

Indeed these terms were not well-defined before using in the manuscript. Now we have added a brief explanation for each term.

- Under the joint action of these forces, allopolyploid subgenomes are further coordinated and degenerated, and subgenomes are often biasedly fractionated" This sentence has some unclear terminology. Does "coordinated" mean co-adapted, co-inherited, or something else? Is "biasedly fractionated" referring to biased inheritance or evolution of one of the parental subgenomes?*

We apologize for not using accurate terms. With “coordinated” we emphasized the evolution of both homoeologs depends on the selection on total expression of both homoeologs, and on both relative and absolute dosages, which may have shifted away from optima after allopolyploidization. “Co-evolved” or “co-adapted” might be a better word.

But the term "biasedly fractionation" has been commonly used for referring to the phenomenon that genes from one subgenome of polyploids are preferentially retained

during diploidization (Woodhouse et al., 2014; Wendel, 2015). Instead of inventing a new term, we prefer to keep the same term for consistency, so readers could link our findings with numerous studies in this field. Now the sentence is changed to “Under the joint action of these forces, allopolyploid subgenomes are further co-adapted and degenerated, and subgenomes are often biasedly retained, termed biased fractionation”.

- *There are a series of paragraphs in the results, starting with "Resynthesized allotetraploids and the natural Cbp had distinct floral morphologies", which consistently reference Figure 1 where they should be referencing Figure 2.*

Thank you for spotting this mistake! Now the numbers have been corrected.

- *'The number of pollen grains per flower decreased in natural Cbp' this wording implies it's the effect of some experimental treatment on Cbp, rather than just measured natural variation.*

Yes, it is not scientifically precise to say this in the Results section, especially when describing details of results. We meant that assuming resynthesized allopolyploids are good approximation of the initial state of natural allotetraploid *C. bursa-pastoris*, our results indicate that the number of pollen grains had decreased in natural *C. bursa-pastoris*. But this is an implication, rather than an observation, so the sentence is better rewritten as “Natural allotetraploids had less pollen grains per flower.”

- *'The percentage of genes showing complete ELD was altogether limited but doubled between resynthesized allotetraploid groups and natural allotetraploids' for clarity, I would suggest revising this to something like "doubled in natural allotetraploids relative to resynthesized allotetraploids"*

Thank you for the suggestion. The sentence has been revised as suggested.

- *I'm not sure I understand what the difference is between expression-level dominance and homeolog expression bias. It seems to me like the former falls under the umbrella of the latter.*

Expression-level dominance and homeolog expression bias are easily confused, but they are conceptually independent. One gene could have expression-level dominance without any homeolog expression bias, or strong homeolog expression bias without any expression-level dominance. The concepts were well explained in Grover et al., (2012) with nice figures.

Expression level dominance compares the total expression level of both homoeologs in allopolyploids with the expression of the same gene in parental species, and judges whether the total expression level in allopolyploids is only similar to one of the parental species. The contributions from different homoeologs are not distinguished.

While homoeolog expression bias compares the relative expression level of each homoeologs in allopolyploids, with no implication on the total expression of both homoeologs.

Let the expression level of one gene in parental species X and Y be $e(X)$ and $e(Y)$, respectively. And let the expression level of x homoeolog (from species X) and y homoeolog (from species Y) in allopolyploids be $e(x)$ and $e(y)$, respectively.

Then a (complete) expression level dominance toward species X means: $e(x)+e(y)=e(X)$ and $e(x)+e(y)\neq e(Y)$;

While a homoeolog expression bias toward species X means: $e(x) > e(y)$, or $e(x)/e(y) > e(X)/e(Y)$, depending on the definition of studies.

Both expression-level dominance and homeolog expression bias have been widely studied in allopolyploids (Combes et al., 2013; Li et al., 2014; Yoo et al., 2014; Hu & Wendel, 2019). As the two phenomena could be in opposite directions, and may be caused by different mechanisms, we think adopting the definitions in Grover et al., (2012) and distinguishing the two concepts would facilitate communication.

- *Is it possible to split up the results in Figure 7 to show which of the two homeologs was lost (i.e. orientalis vs. grandiflora)? Or at least clarify in the legend that these scenarios are pooled together in the figure?*

Maybe using acronyms without explanation made the figure titles hard to understand, but in the original Figure 7 the loss of two homoeologs were shown separately. Figure 7a,c showed the loss of *C. orientalis*-homoeolog (“co-expression loss”), and Figure 7b,d showed the loss of *C. grandiflora*-homoeolog (“cg-expression loss”). Now the legends have been modified to explain the Figure.

- *The paragraph starting with "The extant diploid species" is too long, should probably be split into two paragraphs and edited for clarity.*

The whole paragraph was used to explain why the resynthesized allotetraploids could be a realistic approximation of the early stage of *C. bursa-pastoris* with two arguments:

1. The further divergence between *C. grandiflora* and *C. orientalis* after the formation of *C. bursa-pastoris* should be small compared to the total divergence between the two parental species;
 2. The mating systems of real parental populations were most likely the same as today.
- Now the two arguments were separated as two paragraphs, and the second paragraph has been shortened.

- *On the other hand, the number of seeds per fruit" implies this is evidence for an alternative hypothesis, when I think it's really just more support for the same idea.*

“On the other hand” was used to contrast the reduced number of pollen grains and the increased number of seeds in natural allotetraploids. As both changes are typical selfing syndrome, indeed the two support the same idea. We replaced the “On the other hand” with “Moreover”.

- *'has become self-compatible before the formation" "has become" should be "became".*

The tense of the word has been changed.

- *If natural C. bursa-pastoris indeed originated from the hybridization between C. grandiflora-like outcrossing plants and C. orientalis-like self-fertilizing plants, the selfing syndrome in C. bursa-pastoris does not reflect the instant dominance effect of the C. orientalis alleles, but evolved afterward.' This sentence should be closer to the end of the paragraph, after the main morphological results are summarized.*

Thank you for the suggestion. The paragraph is indeed more coherent after moving the conclusion sentence.

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